

Strategic Heterogeneity in Issue Positioning: Evidence from Congressional Campaigns*

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Abstract

Do candidates maintain consistent left-right issue positions in congressional elections? How do district conditions influence variations in this consistency? While these questions are fundamental to the study of U.S. elections, surprisingly little is known about issue positioning in campaigns. We combine campaign platforms with machine learning methods to estimate the left-right orientations of U.S. House candidates ($n = 4,509$) across six salient issue areas in multiple election cycles (2018-2022). Our validated measures reveal meaningful within-candidate variation in issue positioning that systematically reflects district conditions. Candidates exhibit greater positional heterogeneity in districts that are not safe for their party and where constituent policy preferences are variable. We further demonstrate that this positional variation directly reflects district opinion, as issue-specific positioning aligns with constituency policy preferences. These findings reveal how candidate messaging varies systematically across issue domains and electoral contexts, fundamentally shaping the information environment that structures democratic choice and representation.

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How candidates position themselves on political issues during campaigns carries significant democratic importance, shaping the choices available to voters and establishing benchmarks for representation once in office. In today's highly nationalized electoral environment, it is widely believed that congressional campaign messages reflect the issue positions promoted by the major parties (e.g., Ansolabehere, Snyder, and Stewart 2001; Fiorina 2016; Hopkins 2018; Carson, Sievert, and Williamson 2023). This view implies that candidates advance homogeneous platforms, exhibiting limited capacity to adopt a mixture of left-, right-, and centrist positions across issues. To the extent that such uniformity characterizes campaign communication, it may degrade the quality of democratic representation by privileging partisan consistency over issue-specific flexibility, thereby simplifying voters' electoral choices but at the expense of district-level responsiveness.

Despite its prominence in accounts of contemporary campaigns, this notion that congressional candidates maintain internally consistent left-right positions across issue domains is an assumption rather than an established empirical finding. Existing scholarship has been ill-equipped to test this assumption directly, primarily due to limitations in measurement. While many studies estimate candidates' overall political orientations (e.g., Bonica 2014; Montagnes and Rogowski 2015; Barbera 2015; Case 2027), the unidimensional measures they employ obscure variation in position-taking *across* issue domains (Broockman 2016). Evaluating cross-issue variation in candidates' left-right stances—and whether such variation reflects district-level rather than national considerations—requires issue-level measures of campaign messaging.

We overcome this limitation by developing novel estimates of candidates' left-right positions across six issue areas consistently rated as top public priorities and highly important to vote choice: abortion, education, energy, guns, healthcare, and immigration. These measures allow us to assess the extent to which congressional candidates exhibit positional constraint—whether their left-right positioning on one issue predicts their positioning on all others. We construct these estimates by pairing campaign platforms scraped from candidates' websites ($n = 4,509$, or 75% of all major-party candidates from 2018-2022) with machine-learning methods that extract and scale issue-specific positions from text. Our approach captures the semantic proximity of campaign messaging

to vocabulary associated with the most extreme left- and right-wing positions within each issue domain. We refer to the resulting measures as Candidate Positioning Indexes (CPIs).

We use these measures to examine whether and how candidates' issue-specific positions vary systematically with district-level electoral incentives. We anticipate substantially more cross-issue variability in candidates' campaign messages than prevailing accounts suggest. Although positional constraint confers electoral advantages, we argue that cross-issue variability in positioning can be similarly advantageous under certain conditions. In districts where their own party lacks an electoral advantage, candidates must simultaneously appeal to same-party primary voters and to moderate or opposing-party voters in the general election. We expect candidates who face these dual electoral pressures to pursue a "broad-appeal" strategy, adopting heterogeneous left-right positions to attract a broad range of voters. Similarly, in districts where voters' policy preferences vary markedly across issues, we expect candidates to tailor their positions to local opinion distributions, emphasizing issue-by-issue alignment with constituents.

Our analysis proceeds in three stages. First, we assess within-candidate variation in left-right positioning. We show that nearly half of candidates in our sample adopt a mix of left-, right-, and/or center-leaning positions in their campaign platforms. Consistent with this finding, pairwise correlations among CPIs are modest to weak, indicating substantial positional differentiation across issues. On average, candidates shift 20 percentile ranks in relative positioning from one issue to another—a difference we demonstrate is substantively meaningful using qualitative and quantitative assessments of textual similarity. Second, we examine the district-level conditions under which positional consistency erodes, focusing on partisan competition and policy preference heterogeneity. Results from cross-sectional and two-way fixed effects models align with our expectations: candidates exhibit weaker positional constraint as (1) their districts become less electorally favorable to their party and (2) district-level policy preferences vary more sharply across issue domains. Third, we assess whether this variation corresponds to constituency policy-specific preferences. A series of linear regressions demonstrates that district-level, issue-specific public opinion estimates are associated with relevant CPIs, even after accounting for overall district ideological lean.

Our results underscore the importance of incorporating issue-specific variation into analyses of electoral competition, yielding a more complete account of how candidates adapt their campaign positioning to local electoral environments. We show that candidates' issue-specific positioning aligns with district-level strategic incentives, revealing a degree of local responsiveness in campaign communication that is often presumed absent in today's highly nationalized political era. This correspondence suggests that campaigns can still serve as vehicles for articulating and advancing the policy priorities of heterogeneous local constituencies, thereby reinforcing the representative link between voters and their elected officials. At the same time, however, the heterogeneity we document stands in contrast to the high degree of positional constraint observed in roll-call voting behavior (e.g., Poole and Rosenthal 1997; Marble and Tyler 2022). This divergence between how candidates present themselves during campaigns and how they behave once in office raises fundamental questions about whether campaign platforms function as meaningful commitments to constituents or primarily as strategic tools for winning elections.

Expectations for Positional Constraint Among Elites

A central expectation in contemporary scholarship on American politics is that elites maintain internally consistent positions along a single left-right dimension. Under this assumption, the “ideology” or “political orientations” of politicians meaningfully summarize their policy positions across a wide range of issue areas.¹ This premise fundamentally shapes how scholars model elections, parties, and legislative politics, with politicians' positions typically treated as stable and well organized rather than incoherent or strategically varied (e.g., Jennings 1992; Hetherington 2001; Poole 2007; Levendusky 2009; Bafumi and Herron 2010).

The empirical foundation for this expectation rests primarily on studies of roll-call behavior in the U.S. Congress, with one-dimensional ideal-point models accounting for substantial variation in legislators' vote choices (Poole and Daniels 1985; Poole and Rosenthal 2011) and achieving strong out-of-sample predictive performance (Marble and Tyler 2022). This high degree of posi-

¹We do not assume that politicians' expressed positions necessarily reflect true underlying preferences. Accordingly, we use the terms “political orientations” and “left-right positioning” to describe stated policy stances.

tional constraint is commonly attributed to intense partisan conflict (Theriault 2008), institutional incentives for party unity (Rohde 1991; Cox and McCubbins 2005), and the alignment of economic, social, and cultural issues along the same ideological axis (Layman and Carsey 2002). Yet important exceptions reveal that legislators' policy positions are not uniformly constrained within the legislature (e.g., Heckman and Snyder Jr. 1996; Crespín and Rohde 2010; Jochim and Jones 2013; Moser, Rodríguez, and Lofland 2021). This heterogeneity reflects, in part, how legislators strategically adjust their positions across issues in response to electoral incentives (Yoshinaka and Grose 2011) and institutional arrangements (Roberts and Smith 2003; de Marchi, Dorsey, and Ensley 2021).

Variability in positional constraint observed within Congress raises questions about how candidates navigate issue positioning during electoral campaigns, where institutional constraints are weaker and incentives to tailor messages to constituents are stronger. These differences motivate examining issue-specific electoral messaging as a distinct arena for position-taking. Yet such analysis has been limited by the scarcity of measures that capture candidates' *issue-specific positions*. Existing research has used various approaches to estimate congressional candidates' political orientations, but these approaches typically recover a single summary measure of positioning, reflecting scarcity in available campaign data (e.g., Ansolabehere et al. 2001; Montagnes and Rogowski 2015; Barbera 2015; Bonica 2014; Case 2027).² This reliance on unidimensional measurement has important implications. Unidimensional estimates can capture the consistency of candidates' position-taking, but they cannot identify which issues generate that balance, the extremity or moderation of candidates' positions on those issues, or potential strategic variation that shapes candidates' cross-issue appeals (Aldrich, Montgomery, and Sparks 2014; Broockman 2016).

The evidence reviewed here points to a critical gap in our understanding of campaign politics: existing estimates of candidates' political orientations do not systematically measure issue-specific

²In exception, Moskowitz, Rogowski, and Jr. (2024) estimate left-right issue positions for general election U.S. House candidates (1996-2008) using Project Vote Smart's NPAT survey, but these data are unavailable for recent elections. Bonica and Li (2021) use supervised machine learning to infer issue-specific positions from itemized contributions, yet this approach cannot capture the strategic considerations that candidates themselves use to tailor positions across issues.

stances, limiting our ability to assess both the extent and strategic determinants of cross-issue variation in position-taking. We address this gap by developing new measures of candidates' issue-specific positioning that enable systematic analysis of positional heterogeneity. We situate these measures within a framework that specifies the conditions under which positional heterogeneity is most likely to arise, even in an era of highly nationalized campaigns. Using this approach, we assess whether congressional candidates maintain consistent ideological positions across policy domains or strategically vary their stances in response to electoral incentives.

A Theoretical Framework for Positional Heterogeneity

Why might candidates maintain consistent issue positions rather than vary their stances strategically? In an era of nationalized elections, party labels function as powerful brands that provide ready-made templates for issue positioning. Adopting consistent, party-aligned stances reduces message development costs and voter uncertainty (Kingdon 1973; Snyder and Ting 2002; Lee 2016), while also avoiding the appearance of contradiction or indecision that some voters find undesirable (Powell 1982; Rogowski and Tucker 2018; Milita, Simas, Ryan, and Krupnikov 2017). Positional consistency particularly appeals to primary electorates and organized interests that supply critical campaign resources to candidates (Gimpel, Lee, and Pearson-Merkowitz 2008; Henderson, Sheagley, Goggin, Dancey, and Theodoridis 2022; Porter and Steelman 2023). Together, these dynamics help explain why contemporary scholarship assumes that congressional candidates exhibit high levels of positional constraint, leading voters' local electoral choices to increasingly mirror national party positions.

Yet we argue that, under certain strategic conditions, positional heterogeneity can also be electorally advantageous. We contend that heterogeneous issue positioning is especially advantageous when guided by two key factors: (1) the organization of constituency preferences across policy domains, and (2) the electoral competitiveness of their district.

The Structure of Voter Policy Preferences

Classic research establishes that citizens' policy preferences are weakly constrained (Converse 1964; Kinder 1998), and contemporary work confirms that unidimensional measures capture only limited variation in public opinion (Lauderdale, Hanretty, and Vivyan 2018; Freeder, Lenz, and Turney 2022; Marble and Tyler 2022; Broockman and Lauderdale 2025). This weak constraint reflects substantive heterogeneity in policy preferences: an individual's stance on one issue dimension predicts little about her stance on another, and this cross-issue variation manifests both within and across congressional districts (Broockman 2016; Warshaw and Rodden 2012). Rather than possessing internally consistent political orientations, voters hold distinct preferences across policy domains, creating a complex landscape of public opinion that candidates must navigate strategically.

Candidates must attend to cross-issue heterogeneity in constituency policy opinion because it is closely linked to voting behavior. While citizens value multiple dimensions of representation (Harden 2015), they prioritize policy representation on salient issues (Lapinski, Levendusky, Winneg, and Jamieson 2016). Issue-specific agreement between representatives and constituents influences political approval and vote choice (Ansolabehere and Kuriwaki 2022; Henderson et al. 2022), with some voters even supporting out-partisans who share their policy views (Colao, Broockman, Huber, and Kalla 2025). Most critically, Ahler and Broockman (2018) demonstrates that citizens prefer candidates who align with them on individual issues over candidates with similar overall ideologies. Electoral success, therefore, requires careful attention to voters' policy-specific concerns across distinct issue domains.

Because voters engage in issue-based voting and exhibit heterogeneous left-right policy preferences, candidates face strong electoral incentives to align their own issue positions with these varying preferences. As rational election seekers (Mayhew 1974), candidates must strategically tailor their campaign messages to maximize electoral support. When constituency preferences differ across issue areas, optimal positioning requires issue-specific adjustment rather than adherence to a single ideological location (Downs 1957). Accordingly, greater cross-issue variance in district

preferences should increase incentives for candidates to adopt heterogeneous issue positions. We expect that this strategic adaptation generates systematic within-candidate variation, with candidates' issue-specific left-right positions tracking their district's preferences on those issues.

The Dynamics of District Partisan Competition

Sequential primary and general elections in the United States produce competing incentives for candidate issue positioning. Primary electorates are typically composed of more ideologically extreme (Burden 2004; Nielson and Visalvanich 2017) and politically engaged voters (Brady, Han, and Pope 2007) who hold strong expectations for party-consistent issue representation from candidates (Barber and Pope 2019; Henderson et al. 2022). By contrast, general electorates tend to exhibit greater ideological diversity and contain a higher proportion of moderates and independents (Canes-Wrone, Brady, and Cogan 2002; Hall 2015). These distinctions create cross-pressures in which positions that mobilize primary voters may reduce general election viability, and vice versa.

In districts electorally secure for a candidate's own party, the primary serves as the main arena for electoral competition, with partisan advantages rendering the general election largely non-competitive (Carson, Roberts, and Porter 2020). Winning the primary becomes the candidate's only significant hurdle to office under such conditions, making alignment with primary voter preferences particularly consequential. These safe districts have become increasingly common as the number of marginal seats—where both parties have a reasonable chance of winning—has declined. In same-party safe districts, we expect candidates to maintain relatively greater positional consistency, minimizing departures from party orthodoxy to avoid alienating their partisan base.

In districts where their own party lacks an electoral advantage, candidates face a strategic dilemma: they must appeal sufficiently to their primary electorate to secure the nomination while maintaining broader electability within the general electorate to attain office. Wholesale positional shifts across electoral stages are strategically risky, given constraints imposed by candidates' prior records, the difficulty of credibly signaling changes in position, and the reputational costs of appearing indecisive or opportunistic. Uniform moderation is similarly fraught, as it may alienate committed co-partisans who interpret moderation as ideological compromise. Instead, we expect

candidates running in two-party competitive seats or facing electorally unfavorable environments to adopt greater positional heterogeneity as a strategy to resolve this dilemma. By adopting divergent left-right positions across issues, candidates can simultaneously appeal to their general and primary electorates by tailoring their positioning to specific subconstituencies. Building on existing work, we suspect that this strategy allows candidates to expand their electoral appeal—effectively being “everything to everyone” (Somer-Topcu 2015)—without incurring the costs associated with uniform centrism.

In the sections that follow, we test these expectations by introducing our measurement strategy for estimating candidates’ issue-specific left-right positions, validating these measures, and assessing how district contexts shape patterns of issue representation and cross-issue positioning. In doing so, we provide new evidence on the strategic link between electoral competition, district-level policy opinion, and candidate issue-specific positioning.

Measurement Approach

We measure issue-specific positioning by pairing U.S. House candidate campaign platforms with machine learning methods to scale text along left-right dimensions. Measures are generated for issue areas identified by the Pew Research Center as “very important” to vote choice: abortion, education, energy, guns, healthcare, and immigration. Our measurement framework involves computing word embeddings for all terms in campaign platforms and candidate-issue embeddings for issue-specific text produced in a given election cycle (e.g., Elise Stefanik’s abortion-related platform text in 2018). An embedding is a numerical representation of an object—such as a word or a document—expressed in high dimensional space that captures its semantic meaning. Objects with similar meanings are assigned similar numerical representations. These representations are learned from data based on patterns in how objects occur in relation to one another. As existing work demonstrates (e.g., Tshitoyan, Dagdelen, Weston, Dunn, Rong, Kononova, Persson, Ceder, and Jain 2019; Kozlowski, Taddy, and Evans 2019), the location of embeddings in space captures latent properties of those objects, where embedding angles and magnitudes together determine how strongly and in what way objects express shared latent properties.

We leverage this geometric property of embeddings to place candidate positions for each issue along a left-right dimension using an approach known as semantic projection (see Grand, Blank, Pereira, and Fedorenko 2022 for greater detail). In brief, a candidate’s position on a given issue is estimated by mapping their candidate-issue embedding onto a left-right axis that defines that issue, which produces a continuous measure that captures both the direction and strength of positional alignment. Put differently, semantic projection captures how far a candidate’s issue position lies toward the left- or right-most position for that issue area. We outline the data and methodology underpinning our Candidate Positioning Indexes (CPIs) in greater detail below.

Data: Online Campaign Platforms

Per Druckman, Kifer, and Parkin (2009), an ideal data source on candidate positioning in campaigns will be “unmediated (i.e., directly from the campaign), complete (i.e., covering a full range of rhetorical strategies), and representative of the population of campaigns” (p. 345). However, many commonly used sources of campaign data fail to satisfy these criteria. For example, candidate surveys (e.g., the National Political Awareness Test) provide direct measures of policy stances but typically suffer from low response rates and exclude primary candidates, who are increasingly central to generating electoral competition (Porter and Treul 2025). Data on perceptions of candidates’ positions (e.g., Stone and Simas 2010; Barbera 2015; Bonica 2014) offer broader coverage, but these perceptions may be shaped by non-policy factors.

We employ an original corpus of campaign websites to address these data limitations. Per Porter, Case, and Treul (2025), campaign websites are ubiquitous in modern elections. They present issue-specific positions in the candidates’ own words across multiple policy areas, without time or space constraints, enabling discussion of all issues they consider important to their campaigns. Candidates and their staff spend substantial time crafting website messaging because these sites serve as an informational “hub” in campaigns (Druckman et al. 2009). For this reason, over a dozen states include links to campaign websites on official candidate listings.³

Of the 6,016 ballot-eligible, major-party candidates who ran between 2018 and 2022, we iden-

³As of 2024, these states include AK, AZ, DE, GA, HI, IL, MD, MI, MT, NH, OR, VT, and VA.

tified 5,228 (87%) as having a campaign website. We characterize campaign platforms as the text presented on the “Issues” or “Policy Positions” page of these campaign websites. Teams of research assistants were tasked with locating and extracting platform text from websites via manual downloading. Website text was collected less than two weeks before a candidate’s primary election to maximize data coverage and consistency. We document our data collection strategy in greater detail in Appendix A1. We successfully identified 4,509 candidates as hosting a web-based campaign platform (75% of major party candidates). Candidates without campaign websites tend to be non-competitive challengers (see Porter et al. 2025).

Method: Semantic Projection

Isolating Issue-Specific Text

To produce CPIs for our six issues of interest, we rely exclusively on issue-specific text. Restricting the breadth of text used in our scaling procedure helps to limit noise in measurement (Grimmer and Stewart 2013). To isolate text, we first spliced campaign platforms into individual paragraphs, producing a corpus of 142,835 natural paragraphs.⁴ Next, we randomly sampled roughly 6% of these paragraphs (8,262 total) and labeled them for the presence/absence of content related to our six issue areas of interest.⁵ Full coding instructions can be found in Appendix A2.1.

These labeled paragraphs serve as training data for a series of supervised machine-learning classification models. We trained four classification models for each issue area and employed their predictions to train a stacking ensemble classifier. This approach enabled us to draw on a diverse set of models to achieve better performance, combining each model’s strengths in the final classification. Appendix A2.2 provides greater detail on model training and classifier validation. When classifying a held-out set of paragraphs from campaign platforms, we achieved F1 scores of 0.82 or higher across all models, indicating strong classification performance.

⁴The choice of aggregation at the natural paragraph level follows best practices from Daubler, Benoit, Mikhaylov, and Laver (2012) that emphasize the importance of exogenous unitization of text. We aggregate short paragraphs (i.e., less than fifteen words) into the most proximate multi-sentence paragraph.

⁵Three trained readers labeled paragraphs for issue content. Categorization was not mutually exclusive; multiple issue areas were sometimes discussed within the same paragraph. Of the 8,262 paragraphs, all three coders labeled 305 (4%) to assess inter-coder reliability. For this reliability set, we achieve 87% agreement across all three readers. The distribution of issue prevalence in these labeled paragraphs was relatively consistent across domains (Abortion 3.6%; Education 11.2%; Energy 5.2%; Guns 5.3%; Healthcare 7.9%; Immigration 7.2%).

Estimating Word Embedding Model

We next employ a word embedding model to uncover meaning in these issue-relevant paragraphs. Our estimation approach relies on a Doc2Vec word embedding model. This approach is similar to a traditional word embedding model but allows for the incorporation of covariates during training, producing *covariate-level embeddings* (see Le and Mikolov 2014). In our specification, each paragraph is assigned a covariate that corresponds to the candidate, issue area, and election year.⁶ Resulting covariate embeddings—which we refer to as candidate-issue embeddings⁷—are numerical representations constituting a semantic summary of the rhetorical meaning for *all issue-specific paragraphs* from a given candidate in a given year.

Our Doc2Vec model operates in two parts: the first is similar to a traditional skip-gram Word2vec architecture (Mikolov, Chen, Corrado, and Dean 2013). A word, w_k , is sampled at each iteration, where k is the word’s position in a document. A window of length Δ is extracted twice, once before and after w_k . The words in the window surrounding w_k are our outcomes of interest. These outcomes can be written as $w_\Delta = (w_{k-\Delta}, \dots, w_{k-1}, w_{k+1}, \dots, w_{k+\Delta})$. The model input is an indicator vector, x_k , for the target word, w_k , which selects the corresponding word embedding, β_k . This embedding predicts each word, m , in w_Δ . Figure 1a illustrates this word embedding model architecture.

The second part of our model uses the same architecture: a word, w_k , is sampled at each iteration, and a window of length Δ is extracted before and after. Words falling in this window, w_Δ , represent our outcomes of interest. Instead of using an indicator vector to index the word embedding for target word w_k , the indicator vector, $z_{i,j,t}$, is used to index the candidate-issue embedding, $\zeta_{i,j,t}$, for candidate i on issue area j in election year t . This embedding predicts each word, m , in w_Δ .⁸ Figure 1b illustrates this candidate-issue embedding model architecture.

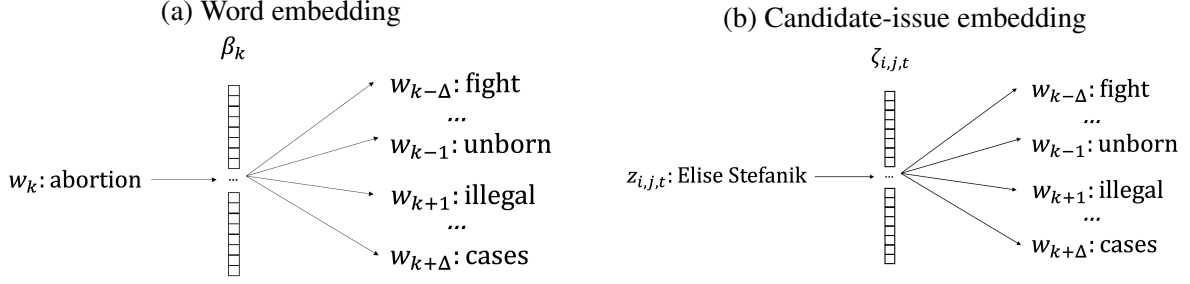
We begin with pre-trained word embeddings from the Google News corpus to set initial model

⁶We classify some paragraphs (2% of all paragraphs) as discussing more than one issue (e.g., a paragraph discussing abortion and healthcare). In these cases, we include duplicate paragraphs in model training.

⁷A more accurate descriptor would be “candidate-issue-year” embedding, as our approach produces unique embeddings for candidates who discuss the same issue across multiple elections. We use “candidate-issue” for brevity.

⁸This process is intuitively similar to taking the average of the word embeddings for candidate i on issue area j in year t . However, the Doc2Vec estimation places less weight on high-frequency words (e.g., the, is, in) that often possess little rhetorical meaning. For this reason, previous work finds Doc2Vec outperforms averaging word embeddings (Lau and Baldwin 2016; Grimmer, Roberts, and Stewart 2022).

Figure 1: Embedding Model Architecture



weights; candidate-issue embeddings are initialized randomly. Our Doc2Vec model alternates between the steps described above during training to tune the embedding parameters. We fit our model using parameter recommendations from Rodriguez and Spirling (2022), including a window of six and an embedding dimension of 300. We employ default hyperparameters from the original Doc2Vec model. We take standard text pre-processing steps for embedding models, as laid out by Rodriguez and Spirling (2022).⁹

Our model produces two sets of outputs. First, our model produces 15,966 word embeddings (β_k)—an embedding for every word in our corpus. Second, our model produces 17,879 covariate embeddings ($\zeta_{i,j,t}$)—an embedding for every candidate-issue-year combination in our data. If our classifier identified no issue-relevant paragraphs for a given candidate in a given year, then we do not generate an embedding for that candidate-issue-year combination. Our modeling approach places covariate and word embeddings in the same dimensional space; we leverage this feature to produce our CPI measures in the following section.

Scaling Candidate Issue Positioning

In this final stage, we scale candidates’ issue positioning along a left-right dimension that defines the discourse for our six areas of interest. Issues are multidimensional, and our measures do not capture positioning along all these dimensions; instead, we pre-specify a dominant cleavage to scale each issue area.¹⁰ Italicized text in Table 1 denotes cleavages of interest.

⁹These pre-processing steps minimally alter the underlying meaning of text (Denny and Spirling 2018). The full list of our steps in text prep-processing is available in Appendix A2.3.

¹⁰Using principal component analysis (PCA) to reduce embedding dimensionality led to a significant decline in CPI validation performance, as the principal components did not always align with task-relevant variance.

Table 1: Issue Cleavages & Positional Endpoint Term Dictionaries

Issue area	Left-most & right-most term dictionaries
Abortion	Pro-Choice (L): reproductive, justice, prochoice, codifying, autonomy, personal... Pro-Life (R): birth, unborn, conception, heartbeat, prolife, ban, born, outlaw...
Education	Federal (L): public, invest, free, tuition, universal, prek, equality, salaries, teachers... Local (R): parents, local, control, decentralize, choice, competition, charter, private...
Energy	Renewables (L): renewable, net, zero, incentives, credits, subsidize, develop... Fossil Fuels (R): oil, coal, deregulate, repeal, free, market, domestic, reserves...
Guns	Restrictions (L): mandatory, background, ban, national, registry, assault, automatic... Access (R): infringed, inherent, unrestricted, repeal, reciprocity, concealed, carry...
Healthcare	Publicly Funded (L): universal, medicareforall, singlepayer, human, right... Privatized (R): free, market, open, competition, choice, consumers, deregulate...
Immigration	Inclusive (L): undocumented, daca, pathway, dreamers, dignity, abolish, ice... Exclusive (R): enforce, law, build, wall, verify, sanctuary, chain, birthright...

Note: The table reports examples of the left-most and right-most term dictionaries used to construct issue-specific Campaign Positioning Indexes. L indicates the left endpoint and R indicates the right endpoint. Entries are abbreviated for presentation; full term dictionaries are available in Appendix A2.4.

To produce our issue-specific measures, we project each candidate’s covariate embedding, $\zeta_{i,j,t}$, onto an axis defined by terms indicative of the left-most and right-most positions for issue j . We define these positional endpoints using term dictionaries and their associated word embeddings. Using dictionaries to capture key quantities of interest is standard in research employing word embeddings (e.g., Kitagawa and Shen-Bayh 2024; Garg, Schiebinger, Jurafsky, and Zou 2018; Garten, Hoover, Johnson, Boghrati, Iskiwitch, and Dehghani 2018). Term selections for these dictionaries were made exogenously from our main corpus of text. In Appendix A2.4, we provide greater detail on our term selection and demonstrate that our estimated CPIs are robust to alternative term selections. Table 1 outlines an abbreviated list of terms in each left/right term dictionary; Appendix Table A2 presents the full dictionaries.

The full scaling procedure for constructing CPIs is as follows. First, for issue j , we average the embeddings for the set of words, R_j and L_j , in our right and left issue dictionaries, respectively. Resulting averages produce two embeddings— ρ_j , and λ_j —indicative of the semantic meaning of the right-most and left-most positions for the cleavage defining j . Second, we subtract λ_j from ρ_j to create an axis of left-right positioning, α_j . The extant literature establishes that applying vector

algebra to word embeddings yields a semantic axis of meaning (e.g., Bolukbasi et al. 2016; Garten et al. 2018; Kozlowski et al. 2019; Grand et al. 2022). In our application, this axis represents the left-right dimension for the cleavage that defines issue j . Finally, for each candidate i in year t with an estimated covariate embedding for issue j , we project $\zeta_{i,j,t}$ onto α_j . This is achieved by taking the magnitude of $\zeta_{i,j,t}$ and multiplying it by the cosine similarity between $\zeta_{i,j,t}$ and α_j . As resulting values increase (decrease), the candidate’s text is placed closer to the right-leaning (left-leaning) end of the issue-specific semantic axis, with larger absolute values indicating greater distance along that dimension. This estimation is formally written as:

$$\begin{aligned}
R_j &= \{w_1, w_2 \dots w_n\} \\
L_j &= \{w_1, w_2 \dots w_n\} \\
\rho_j &= \overline{\beta_{k \in R_j}} \\
\lambda_j &= \overline{\beta_{k \in L_j}} \\
\alpha_j &= \rho_j - \lambda_j \\
CPI_{i,j,t} &= \|\zeta_{i,j,t}\| \cos(\alpha_j, \zeta_{i,j,t})
\end{aligned}$$

CPI Measurement Validation

The Appendix reports a series of validation exercises assessing the semantic and predictive validity of the Candidate Positioning Indexes (CPIs). First, in Appendix Section A3, we evaluate convergent validity by comparing CPI estimates to human ratings of campaign platform text. Given humans’ superior ability to read and understand natural language, directly comparing the quantitative scaling of text with qualitative human judgments provides a meaningful benchmark for semantic validity. Across six issue areas, CPI scores closely correspond with average human ratings of text on a left-right scale, with correlations ranging from 0.74 to 0.88. Instances of ambiguity or irrelevance identified by human readers are rare, suggesting that the automated scaling captures substantively meaningful positional differences rather than noise.¹¹

¹¹To further diagnose measurement quality, we conduct an exploratory factor analysis of CPIs. In an era of high polarization, we would expect CPIs to collapse onto a single dimension. Measures dominated by error are unlikely to cohere around a common latent dimension. Instead, we find strong evidence of unidimensional structure: all CPIs load heavily on a single factor (loadings range from 0.74 to 0.89), which accounts for 64 percent of the total variance.

Second, in Appendix Section A4, we assess predictive validity by examining whether issue-specific CPI extremity is associated with contributions from Political Action Committees (PACs) for position-aligned groups. Across abortion, energy, and gun policy domains, candidates whose issue positions are more extreme in the direction favored by aligned PACs are more likely to receive contributions, even after controlling for general ideological positioning. These patterns are consistent with existing theories of interest group behavior and provide further evidence that CPIs capture politically relevant dimensions of candidate positioning.

Measuring Positional Constraint with CPIs

We use our Candidate Positioning Indexes (CPIs) to measure the degree of positional constraint in congressional candidates' campaign platforms. Because positional constraint is measured based on expressed issue positions, our analysis necessarily excludes candidates who do not address any relevant issues in their campaign platforms (26% of all major-party candidates)¹² as well as candidates who address only a single relevant issue (4% of all major-party candidates). Notably, approximately half of all candidates have CPIs available for at least four policy areas.

Given that CPIs are derived from a fixed, issue-specific semantic axis, they enable comparability over time within individual issue domains (e.g., abortion positioning from 2018 to 2022) but do not facilitate direct cross-domain comparisons (e.g., abortion and immigration positioning in 2018). Producing measures that are directly comparable across issue domains would require the development of calibrated dictionaries, in which a CPI value for a particular issue corresponds to an equivalent semantic position across all other issue-specific CPI scales. While theoretically possible, this approach is implausible in practice given heterogeneity in issue-specific language, salience, and ideological structures. This limitation exemplifies a more pervasive challenge associated with developing continuous measures from observational political data. Such concerns have been extensively discussed in the spatial modeling literature, including critiques of the use of NOMINATE-based ideal-point estimates in cross-temporal applications (e.g., Roberts 2007; Mc-

Standard fit statistics further support a one-factor solution (TLI = 0.99; RMSEA = 0.045, 90% CI [0.038, 0.052]).

¹²This category also includes candidates who do not have a publicly available campaign platform.

Carty 2016; Caughey and Schickler 2016). In light of this limitation, we conceptualize positional constraint relationally—focusing on the internal consistency of candidates’ relative placements across CPIs rather than on the interpretation of raw CPI values.

We employ two approaches to convert CPIs into relative quantities. First, we transform CPIs into percentile ranks, ordering candidates who took a position on each issue from lowest (most left) to highest (most right). Cross-issue comparisons using this measure assume that equivalent ranks reflect comparable degrees of extremity—effectively treating a candidate at the 80th percentile on abortion as equally extreme as one at the 80th percentile on immigration. Moreover, this approach assumes that ordinal information captures all politically meaningful variation, implicitly discarding information about the magnitude of positional differences between candidates.

A limitation of this assumption is that percentile-ranked CPIs are most informative when there is meaningful within-issue variation in candidates’ positions. When candidates are tightly clustered within an issue domain, small differences in estimated CPI scores may generate relatively large differences in percentile rank, making rankings more sensitive to measurement error. In the limiting case where all candidates from a party take the same true position on an issue, percentile ranks would consist entirely of measurement error. Accordingly, comparisons of positional constraint across issue pairs are contingent on relatively comparable levels of within-issue variance: low-variance issues may produce weaker apparent constraint because ranks contain less substantive information, while higher-variance issues may yield stronger apparent constraint because rankings better recover meaningful positional differences. Empirically, this concern is mitigated in our application, as within-party standard deviations of raw CPIs are relatively similar across issues, ranging from 0.36 to 0.43 among Democrats and from 0.34 to 0.45 among Republicans.

As a secondary approach, we transform CPIs into a trichotomous measure, assigning values of left, centrist, and right. This transformation standardizes positions within each issue by classifying texts according to whether their CPI scores fall above (right), below (left), or near the issue-specific mean. By emphasizing coarse positional categories rather than continuous variation, the trichotomous measure is less sensitive to differences in scale dispersion or calibration across is-

Table 2: Candidate Positional Constraint Across Binning Thresholds for Trichotomized CPIs

Binning threshold	Positionally consistent	Positionally inconsistent	
		Left/right-centrist	Cross-spectrum
Within 0.05 SD of mean	0.517	0.073	0.410
Within 0.10 SD of mean	0.490	0.135	0.375
Within 0.25 SD of mean	0.391	0.334	0.275
Within 0.33 SD of mean	0.344	0.430	0.226
Within 0.50 SD of mean	0.256	0.599	0.145

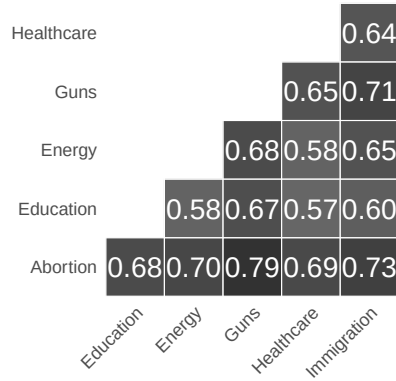
Note: Positional constraint identifies when all positions fall within a single category (left, right, or centrist); left/right-centrist identifies when positions span the left or right category and the centrist category; and cross-spectrum identifies when positions span both left and right or all three categories. Column categories are mutually exclusive. CPI refers to Campaign Positioning Index. SD refers to standard deviation.

sues. Compared to percentile-ranked CPIs, this approach relies on weaker ordinal assumptions but necessarily sacrifices information about within-category heterogeneity (e.g., distinctions between standard partisan and more extreme positions). Cross-issue comparisons using this trichotomous measure still require assuming that coarse categories carry comparable substantive meaning across issue domains, despite differences in underlying distributions and semantic content. The trichotomous transformation further entails defining issue-specific category thresholds, which require an explicit decision about where to draw the boundary between categories.

In Table 2, we illustrate how alternative binning thresholds affect conclusions about the pervasiveness of positional constraint in campaign messaging. We define binning thresholds using issue-specific distances from CPI means, measured in standard deviation units. For example, under a 0.25 SD threshold, candidates with CPI-Abortion scores at least 0.25 standard deviations above (below) the issue-specific mean are classified as taking right-leaning (left-leaning) positions. The positional constraint category in Table 2 identifies candidates whose trichotomized CPIs all align with a single political orientation (left, right, or centrist). The left/right-centrist category identifies positionally inconsistent candidates whose CPIs span one political orientation (left or right) as well as the centrist category. The cross-spectrum category identifies positionally inconsistent candidates whose CPIs span both the left and right categories or all three political orientations.

As binning thresholds widen, positional consistency declines mechanically, left/right-centrist

Figure 2: Percent Agreement on Trichotomized CPIs (Left, Center, Right)



inconsistency becomes more prevalent, and cross-spectrum inconsistency becomes less prevalent. Notably, even under a highly restrictive threshold of 0.05 SD, only slightly more than half of the candidates in our sample are classified as positionally consistent. Because our primary objective is to interrogate the assumption of high positional constraint in campaigns, we favor a measurement specification that errs on the side of overstating positional consistency while also avoiding implausibility in the distribution of positionally inconsistent candidates.¹³ For this reason, we adopt the 0.25 SD threshold for our subsequent analyses (bolded in Table 2).¹⁴

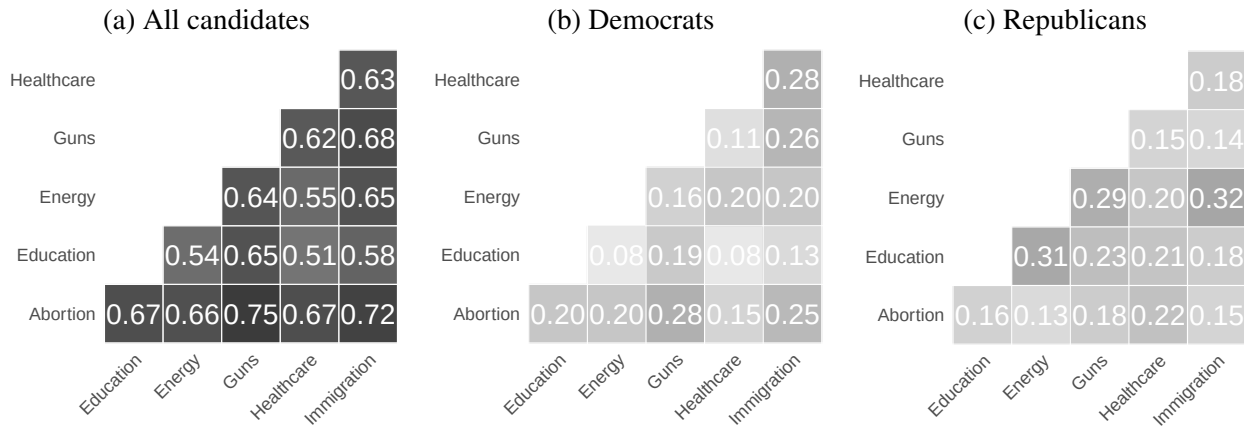
Results

The empirical analyses that follow provide support for our theoretical expectations. First, we show that candidates exhibit significant heterogeneity in their left-right positioning, and these differences reflect meaningful shifts in their positional extremity across issue domains. Second, positional variability increases as a candidate's district (1) becomes less favorable towards her own party, and (2) exhibits greater cross-issue variance in policy opinion. Third, this variation suggests constituency responsiveness: issue-specific public opinion at the district level predicts issue-specific candidate positioning even after accounting for overall district ideology.

¹³For instance, under the 0.05 SD binning threshold, nearly *half* of all candidates exhibit cross-spectrum positional inconsistency, an unlikely descriptive in our current era of high partisan polarization.

¹⁴We do not interpret the resulting proportions as literal estimates. Instead, the 0.25 SD threshold reflects a trade-off between overstating positional consistency and avoiding implausible inflation of cross-spectrum inconsistency.

Figure 3: Percentile-Rank Pairwise CPI Correlations, By Candidate Type



Establishing Heterogeneity in Issue-Level Positioning

We begin by assessing the extent to which candidates adopt consistent left–right positions across issue domains. Table 2 indicates that positional constraint is less pervasive than contemporary accounts of congressional campaigns often imply. To investigate this pattern further, Figure 2 disaggregates positional agreement by issue pair, summarizing how often candidates adopt the same categorical stance (left, right, or centrist) across two issue domains. Agreement levels vary substantially: some issue pairs—most notably guns and abortion—exhibit relatively high agreement, while others—such as energy and education—show markedly lower agreement.

Beyond this variation in categorical alignment, candidates also differ in the intensity of their left-right positioning within political orientations. Figure 3 examines how candidates’ positions covary across issue domains using percentile-ranked CPIs.¹⁵ Pooling all congressional candidates (left panel), pairwise correlations indicate moderate cross-issue alignment, with most falling below 0.70—though this partly reflects between-party differences in positioning. Disaggregating by party (middle and right panels) attenuates these correlations, with most falling below 0.40. These patterns indicate that party affiliation structures broad directional differences but does not tightly anchor candidates’ relative left-right positions across issue domains.

Building on the correlational patterns above, we next quantify the magnitude of within-candidate

¹⁵Figure A4 depicts identical takeaways for pairwise correlation coefficients with raw CPI scores.

Table 3: Average Absolute Differences in Percentile Rankings for Candidate-Level CPI Pairings

	Abortion	Education	Energy	Guns	Healthcare	Immigration
Abortion	0	19.68 [7.75, 28.77]	19.60 [6.84, 28.60]	16.51 [6.08, 24.06]	19.54 [7.23, 29.08]	17.47 [6.38, 25.50]
Education	19.68 [7.75, 28.77]	0	21.85 [7.98, 32.01]	20.04 [7.25, 29.94]	22.30 [8.34, 32.77]	21.78 [8.37, 31.57]
Energy	19.60 [6.84, 28.60]	21.85 [7.98, 32.01]	0	20.36 [7.89, 28.97]	21.19 [7.67, 31.97]	20.18 [7.21, 29.05]
Guns	16.51 [6.08, 24.06]	20.04 [7.25, 29.94]	20.36 [7.89, 28.97]	0	21.17 [7.55, 31.08]	18.28 [6.74, 26.45]
Healthcare	19.54 [7.23, 29.08]	22.30 [8.34, 32.77]	21.19 [7.67, 31.97]	21.17 [7.55, 31.08]	0	20.32 [7.19, 29.46]
Immigration	17.47 [6.38, 25.50]	21.78 [8.37, 31.57]	20.18 [7.21, 29.05]	18.28 [6.74, 26.45]	20.32 [7.19, 29.46]	0

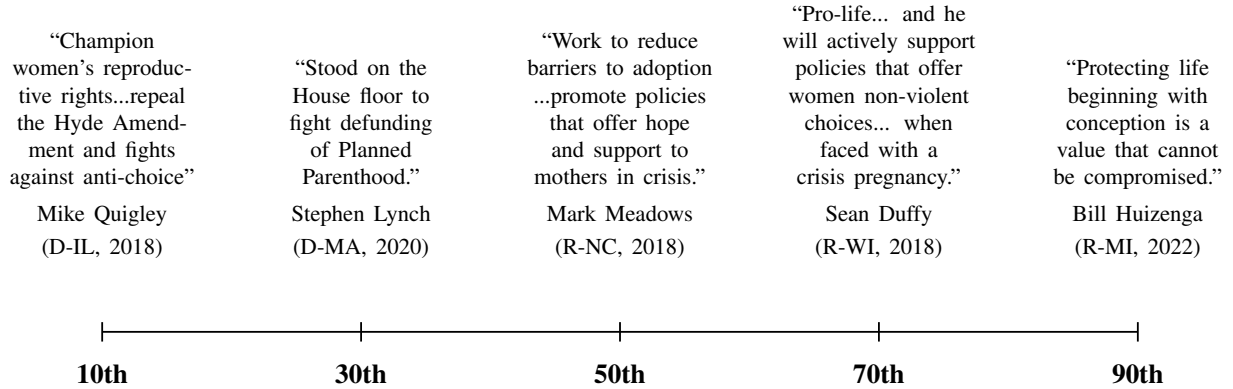
Note: The table reports the average absolute difference in candidates' percentile-ranked Campaign Positioning Indexes (CPIs) across pairs of issue areas. Values are calculated among candidates with non-missing CPI scores for both issue areas in a given pairing. Bracketed values report the 25th and 75th percentiles of the absolute differences. Larger values indicate greater movement in candidates' relative positioning across pairs of issue areas.

variation in positioning across issue pairs. For candidates discussing both issues in a given pairing, we calculate the absolute difference in their CPI percentile rankings and average these differences to characterize typical cross-issue positional shifts. Table 3 reports these magnitude differences in percentile rankings, with each cell presenting the mean absolute difference and interquartile range. The results reveal substantial positional variation: on average, candidates shift by approximately 20 percentile ranks in their left-right positioning across issue domains. The interquartile ranges indicate meaningful dispersion, with typical shifts ranging from 8 to 30 percentile points.

To contextualize the substantive magnitude of 20-percentile shifts in positional rank, we present illustrative excerpts from campaign platforms at corresponding intervals of CPI-Abortion.¹⁶ Figure 4 reveals clear differences in candidate position-taking across the distribution, ranging from strong pro-choice advocacy at the 10th percentile (left) to explicit pro-life positioning at the 90th percentile (right). The substantive differences associated with these incremental shifts are evident in specific candidate comparisons: Mark Meadows (R-NC, 50th percentile) emphasizes “reducing barriers to adoption,” representing a centrist position focused on alternative solutions to

¹⁶Complete textual comparisons across all CPI issue areas are available in Appendix Tables A9 through A14.

Figure 4: Campaign Platform Excerpts at 20 Percentile-Rank Intervals for CPI-Abortion



abortion, while Sean Duffy (R-WI, 70th percentile) adopts a more conservative stance by taking a pro-life position will still wanting women to have “choices and opportunities when faced with a crisis pregnancy.” This contrast illustrates how 20-point shifts in CPI-Abortion correspond to discernible changes in substantive positioning, with distinct policy emphases emerging at different points along the left-right spectrum.

We assess these textual shifts more systematically in Table 4 using a series of keyness analyses comparing abortion-related campaign statements across adjacent CPI-Abortion percentile bands.¹⁷ Keyness is a corpus-linguistic approach that identifies terms that occur significantly more frequently in one set of texts than in another. By comparing adjacent percentile groupings (e.g., 90th vs. 70th, 70th vs. 50th), we isolate the incremental textual shifts associated with movement along the left-right spectrum and evaluate whether a 20-point change in percentile rank on CPI-Abortion reflects meaningful differences in how candidates articulate their positions. Keyness analysis results for all other CPI issue areas are included in Appendix Tables A15 to A19.

Table 4 displays the top twenty discriminating terms for each percentile comparison. Candidates in higher-percentile groups more frequently employ language such as *life*, *unborn*, *conception*, and *pro-life*, while lower-percentile candidates emphasize terms related to *healthcare*, *access*, and *rights*. If 20-point percentile rank differences in campaign platform text merely reflected mea-

¹⁷Each percentile group includes candidates within a 10-point range of the cutoff (e.g., the “90th percentile” group includes candidates ranked from the 85th to 95th percentiles), ensuring sufficient sample sizes while maintaining an average 20-point gap between groups.

Table 4: Abortion Word Exclusivity Comparison: CPI-Abortion

Percentile Comparison	Higher Percentile Right-Leaning	Lower Percentile Left-Leaning
90th vs. 70th	life, liberty, vote, baby, happiness, precious, year, gift, conception, every, alive, babies, pursuit, endorsed, among, strong, throughout, world, killing, first	tax, ability, values, nations, education, crisis, good, personal, centers, family, action, access, pro-life, hyde, pay, caucus, planning, families, incest, texas
70th vs. 50th	life, unborn, conception, h.r, begins, congressman, pro-life, protection, act, nations, baby, innocent, tax, god, protect, human, sanctity, taxpayer, protecting, cosponsored	women, access, reproductive, health, womens, control, healthcare, choose, services, care, law, bodies, maternal, including, affordable, decision, rights, quality, roe, pregnancies
50th vs. 30th	pro-life, life, unborn, children, abortion, conservative, pursuit, possible, values, babies, position, abortions, mother, lives, pregnancies, liberty, happiness, illegal, child, can	reproductive, womens, health, access, equal, womans, care, sexual, decisions, safe, planning, contraception, services, lgbtq, justice, information, screenings, economic, family, coverage
30th vs. 10th	life, defund, religious, stand, republican, attempt, beliefs, doctor, bill, now, say, body, like, dollars, deny, planned, adoption, family, year, parenthood	care, abortion, access, reproductive, patients, accessible, girls, get, affordable, face, mental, hyde, ensure, equality, much, repeal, doctors, must, ability, still

Note: The table reports words that are most distinctive of texts near each CPI-Abortion percentile comparison. Higher percentile comparisons identify words more common among candidates in the higher, more right-leaning percentile band relative to candidates approximately twenty percentile points lower. Lower percentile comparisons identify words more common among candidates in the lower, more left-leaning percentile band relative to candidates approximately twenty percentile points higher. Terms are identified using chi-squared keyness statistics from candidate-level abortion text. Comparable word exclusivity comparisons for the remaining CPI issue areas are reported in Appendix Tables A15–A19.

surement noise or arbitrary variation, we would expect these comparisons to yield diffuse or non-specific language. Instead, the systematic emergence of politically coherent terminology indicates that observed positional shifts represent genuine differences in candidates’ left-right positioning across this issue area. Taken together, these results reveal substantively large and politically meaningful patterns of heterogeneity in how candidates articulate issue-specific positions.

Strategic Determinants for Positional Heterogeneity

Having established descriptive heterogeneity in congressional candidates’ issue-specific positions, we turn to evaluating the district-level strategic determinants of this variation. We first examine district electoral competition, expecting positional heterogeneity to be most evident among candidates running in electorally competitive or unfavorable districts. In such contexts, adopting

divergent left-right stances across issue domains may help a candidate to broaden her electoral appeal. Second, we expect greater positional heterogeneity among candidates running in districts with higher variation in public opinion across issue domains. Consistent with Fenno's (1978) observation that legislators navigate heterogeneous policy demands, substantial variation in district opinion should encourage candidates to adjust their positions across issues to accommodate diverse constituent preferences. Examining these dynamics clarifies whether candidates' positional variability systematically reflects district-level electoral incentives.

Although prior work has examined variation in candidates' issue positions, our analysis differs in both theoretical focus and methodological approach. Existing studies primarily evaluate the electoral effects of positional inconsistency while relying on indirect and perceptual proxies to capture positional constraint in campaign communication (e.g., Rogowski and Tucker 2018; Somer-Topcu 2015). In contrast, we directly estimate candidates' stated positions across multiple issue domains, allowing us to more precisely estimate the magnitude and structure of positional constraint. Moreover, our investigation emphasizes the strategic factors underlying positional heterogeneity, focusing on how electoral contexts shape candidate communication rather than on the electoral consequences of this positioning strategy.

We recognize that our theoretical framework considers district preferences primarily as exogenous to candidate positioning. However, the causal dynamics are likely bidirectional: while candidates may adjust their positions to reflect constituency opinions, campaign activities can also influence district preferences over time. Therefore, our analyses aim to describe systematic associations between district characteristics and candidate positions, rather than establish causality. We interpret the findings within this contextual limitation.

To examine the relationship between candidate positional heterogeneity and district conditions, we estimate a series of regression models. We operationalize positional heterogeneity using two outcome measures. First, we calculate within-candidate standard deviations for percentile-ranked CPIs; higher values indicate greater cross-issue variation in a candidate's stated positions. Second, we construct a binary indicator for whether (1) or not (0) a candidate adopts a mixture of left, right,

Table 5: District-Level Policy-Specific Survey Questions

Issue Area	Survey Question: Do you support or oppose...?
Abortion	<i>Pro-Choice</i> (L): Always allowing a woman to obtain an abortion as a matter of choice <i>Pro-Life</i> (R): Make abortions illegal in all circumstances
Education	<i>Local</i> (R): Provide tax-funded vouchers to be used for private or religious schools
Energy	<i>Renewables</i> (L): Halt new oil and gas leases on federal lands <i>Fossil Fuels</i> (R): Increase fossil fuel production in the U.S.
Guns	<i>Restrictions</i> (L): Banning assault rifles <i>Restrictions</i> (L): Require criminal background checks on all gun sales
Healthcare	<i>Publicly Funded</i> (L): Providing Medicare (Medicare-for-all) for all Americans <i>Privatized</i> (R): Repealing the entire Affordable Care Act
Immigration	<i>Exclusive</i> (R): Charging immigrants who enter the U.S. illegally with a federal crime <i>Exclusive</i> (R): Deport all undocumented immigrants

Note: The table lists the district-level survey questions used to construct the policy-specific public opinion measures. L indicates survey items aligned with the left endpoint of the issue scale, and R indicates survey items aligned with the right endpoint. Items are recoded as needed so that higher values correspond to more right-leaning district opinion.

and centrist positions across issues, based on our trichotomized CPI measure. We estimate OLS models for the continuous outcome and logistic regressions for the binary outcome.

Our main explanatory variables capture district-level electoral competition and cross-issue variation in district policy opinion. We measure electoral competition using the two-party vote share received by the candidate's copartisan in the most recent presidential election.¹⁸ We expect greater opposite-party favorability to be associated with greater positional heterogeneity among candidates. We measure variation in district policy opinion using estimates from de Benedictis-Kessner, Greenwood, Stephanopoulos, and Warshaw (2024), which draw on policy preference questions from 18 large-scale surveys to estimate public opinion at the congressional district level. These estimates are produced using multilevel regression with poststratification (MRP), which models individual survey responses as a function of demographic and geographic characteristics and then poststratifies to district populations. Table 5 reports the wording of questions used in our analyses. We aggregate opinion estimates for these questions to construct summary measures at the issue-

¹⁸Specifying district competitiveness as a trichotomized categorical variable or including a quadratic term yields substantively identical results.

district-year level.¹⁹ To capture variation in district policy preferences, we percentile-rank these summary measures within each policy area and compute, for each district, the standard deviation of the resulting ranks.²⁰

We estimate two sets of models for each outcome variable. The first specification includes candidate and year fixed effects. Candidate fixed effects account for unobserved, time-invariant politician characteristics, while year fixed effects capture election-specific shocks common to all candidates. The second specification consists of cross-sectional models that include controls for candidate prior political experience, partisanship, gender, fundraising, and other district-level characteristics. In all logistic regressions, we additionally control for the number of issue positions a candidate takes, since candidates who articulate more positions are mechanically more likely to exhibit positional inconsistency.

Table 6 presents results from these regression models. We find that candidates running in districts with greater opposing-party favorability exhibit higher levels of positional heterogeneity. In Models 1 and 2, the coefficient for *Other-Party Presidential Vote Share* is positive and statistically significant, indicating that as other-party electoral strength increases, candidates adopt more variable positions across issues. Models 3 and 4 show a similar pattern, with higher other-party favorability associated with a greater likelihood of positional inconsistency. These results suggest that electoral insecurity weakens positional constraint, consistent with our expectations.

We also find that cross-issue variation in district policy opinion is associated with greater positional heterogeneity. In Models 1 and 2, the coefficient for *District Policy Opinion Variation* is positive and statistically significant, indicating that candidates running in districts with more heterogeneous average policy preferences adopt more variable positions across issues in their campaign platforms. Model 4 yields a consistent result, with greater district opinion variation associated with a higher likelihood of positional inconsistency. In Model 3, the estimated effect is in the

¹⁹Component estimates are first reoriented to ensure consistent directionality, then rescaled to have a mean of zero and a standard deviation of one, and finally combined by addition with equal weighting.

²⁰We adopt this approach because candidates address different subsets of issues in their campaign platforms. Percentile ranks ensure comparability across candidates and mitigate concerns that variation in issue coverage drives the results. Replicating the analysis using variance based on raw opinion estimates yields substantively identical results.

Table 6: Candidate positional heterogeneity and district-level conditions

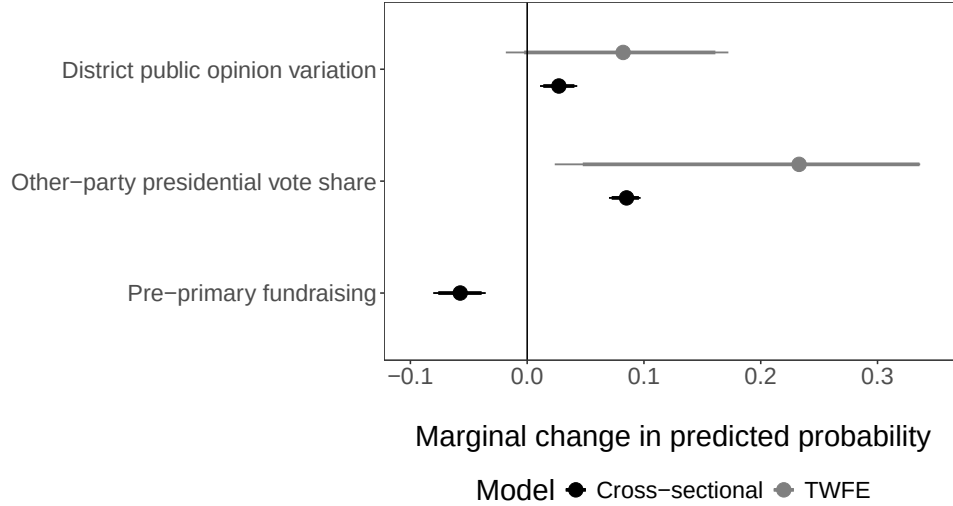
	Standard deviation of indexes		Positional inconsistency	
	(1)	(2)	(3)	(4)
Other-party presidential vote share	0.0008* (0.0004)	0.0006** (0.0001)	0.1203** (0.0434)	0.0244** (0.0027)
District policy opinion variation	0.1274 [†] (0.0660)	0.0232 [†] (0.0137)	4.8243 (3.0654)	1.1186** (0.3438)
Republican		−0.0161** (0.0029)		−0.4790** (0.0765)
Experienced challenger		−0.0007 (0.0038)		−0.2462* (0.1044)
Incumbent		0.0141** (0.0045)		0.3504** (0.1155)
Female candidate		−0.0001 (0.0030)		−0.1118 (0.0820)
Contested primary		−0.0018 (0.0040)		−0.0676 (0.1065)
Partisan primary		−0.0071 (0.0044)		−0.2737* (0.1088)
Pre-primary fundraising (logged)		−0.0006 [†] (0.0003)		−0.0487** (0.0090)
Redistricted		0.0057 (0.0153)		−0.4849 (0.7902)
Constant		0.1577** (0.0095)		−0.4111 (0.2805)
Observations	4,119	4,119	4,119	4,119
Candidate fixed effects	Yes	No	Yes	No
Year fixed effects	Yes	Yes	Yes	Yes
Controls for number of positions	No	No	Yes	Yes

Note: The standard deviation measure is calculated using percentile-ranked Candidate Positioning Indexes, and positional inconsistency is based on trichotomized Candidate Positioning Indexes. Models 1–2 are estimated using linear specifications, and Models 3–4 are estimated using logistic specifications. Standard errors are clustered at the candidate level. [†] $p < .10$; * $p < .05$; ** $p < .01$.

expected direction but does not reach conventional levels of statistical significance.

To clarify the substantive magnitude of estimated coefficients, Figure 5 presents marginal changes in the predicted probability of positional inconsistency (Table 6, Models 3–4). Marginal effects are calculated as the change in the predicted probability associated with a one-standard deviation shift—from one-half standard deviation below to one-half standard deviation above the covariate mean—holding all other covariates at observed values. A one-standard deviation in-

Figure 5: Marginal Change in Predicted Probabilities for Positional Inconsistency



Note: Predicted probabilities are generated from the models reported in Columns 3–4 of Table 6. Marginal changes reflect a shift from one-half standard deviation below to one-half standard deviation above the mean of each covariate. Error bars indicate 95% and 90% confidence intervals. TWFE refers to two-way fixed effect model.

crease in *Other-Party Presidential Vote Share* corresponds to a nine-percentage-point increase in the probability of positional inconsistency in the cross-sectional model and a 23 percentage-point increase in the two-way fixed effect model. The effect of *District Policy Opinion Variation* is more modest: a one-standard-deviation shift corresponds with a four-percentage-point increase in the cross-sectional model and an eight percentage-point increase in the fixed effect model.

As a point of comparison, we next estimate the marginal change in the predicted probability of positional inconsistency associated with a one-standard deviation increase in *Pre-Primary Fundraising*.²¹ We use this covariate as a benchmark because fundraising is a well-established indicator of campaign quality and professionalization, capturing candidates' capacity to mobilize resources and build organizational infrastructure prior to the primary election (e.g., Herrnson 1992; Thomsen 2022). Better-resourced campaigns tend to deploy more carefully calibrated messages (Porter, Treul, and McDonald 2024), making fundraising a useful reference point for evaluating the relative importance of district-level factors in shaping positional constraint. The marginal effect of *Other-Party Presidential Vote Share* exceeds that of an equivalent increase in *Pre-Primary*

²¹All one standard deviation increases are calculated by going from a half standard deviation below the variable's mean to a half standard deviation above the variable's mean to ensure comparability of marginal changes in predicted probability.

Fundraising, while the marginal effect of *District Policy Opinion Variation* is comparable in magnitude. These results underscore that district-level electoral and opinion environments are substantively meaningful determinants of candidates' positional strategies.

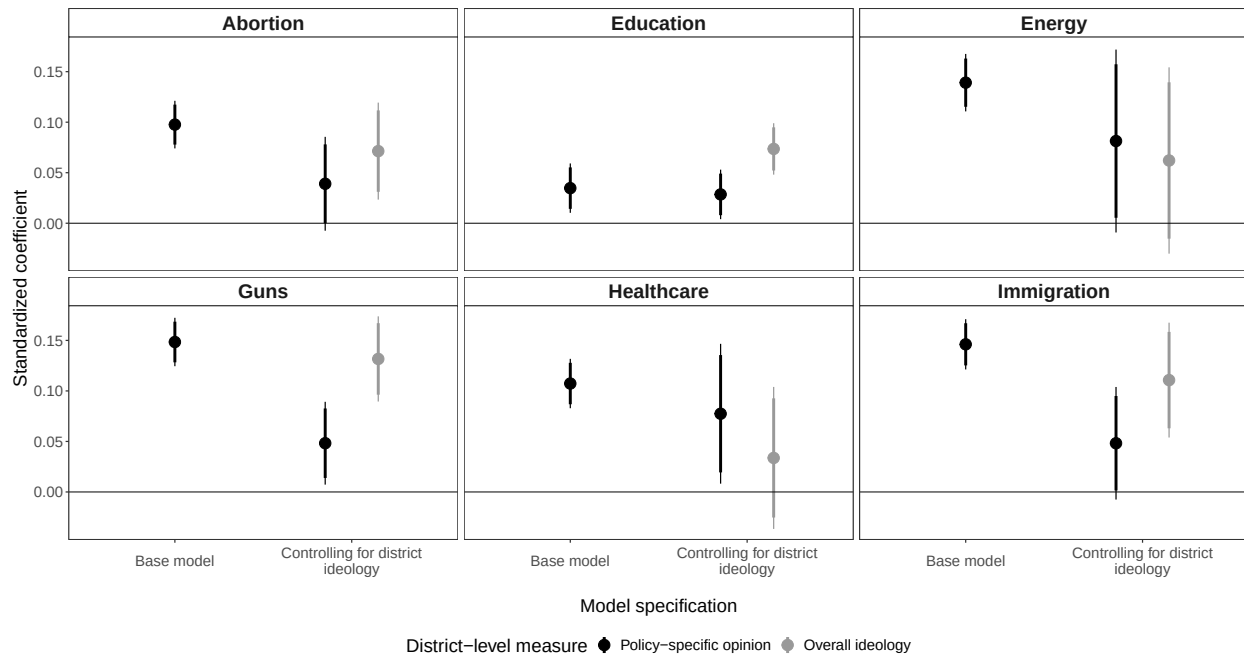
In sum, we find that candidate positional constraint in campaign communications varies systematically with district-level electoral and opinion environments. Across various model specifications and outcome measures, candidates exhibit greater positional heterogeneity when they face electorally unfavorable districts and when average public opinion within the district is more varied. This pattern points to a close association between candidates' cross-issue positioning and issue-specific district preferences, an association we examine more directly in the next section.

District Policy-Specific Opinion & Candidate Issue Positioning

Prior research shows that voters and constituencies often hold inconsistent left-right preferences across issues (Warshaw and Rodden 2012; Broockman 2016). In such contexts, candidates seeking to maximize electoral appeal have incentives to tailor their issue-specific positions to this policy-specific heterogeneity in district preferences. For instance, a candidate running in a district that is right-leaning on abortion but left-leaning on environmental policy would adopt positions that reflect these distinct configurations rather than adhere strictly to the district's overall ideological profile. This pattern of issue-specific alignment is consistent with the relationship between district-level opinion variability and positional heterogeneity documented in the previous section.

To investigate this relationship, we estimate a series of linear regressions where the outcome variable is a candidate's issue-specific CPI. The main explanatory variable is district policy opinion, operationalized using the issue-district-year summary measures described in the previous analysis. We once again control for relevant candidate and district characteristics. If candidates' issue positioning reflects district policy preferences, we should observe a statistically significant, positive relationship between public opinion estimates and issue-relevant CPIs. In a second set of analyses, we reestimate these models while controlling for overall district ideology, using ideal-point estimates from Warshaw and Tausanovitch (2022). This allows us to evaluate whether candidates' issue-level positions align with policy-specific public opinion beyond what would be expected

Figure 6: OLS Regression Model Coefficients: District Policy-Level Opinion and CPIs



Note: Panels display model coefficients. The left coefficients are drawn from base models (Appendix Table A20). The right coefficients are drawn from models replicating these analyses, controlling for district overall ideological orientation (Appendix Table A21). Error bars indicate 95% and 90% confidence intervals.

from their district's overall ideological leaning.

Figure 6 plots coefficients from linear regression models predicting candidate CPIs, with each panel corresponding to a different issue domain. All covariates are standardized to have a mean of zero and standard deviation of one, allowing coefficient magnitudes to be compared directly within each model. Within each panel, black points report estimated coefficients for district-level, policy-relevant public opinion, and gray points report coefficients for overall district ideology. Estimates from the left-hand models are drawn from baseline specifications that include candidate- and district-level controls, while estimates from the right-hand models additionally control for the district's overall ideological orientation. Full model results are reported in Appendix Tables A20 and A21. Error bars indicate 95% and 90% confidence intervals.

Figure 6 shows a consistent association between district-level policy preferences and candidates' issue-specific CPIs. Across all six baseline model specifications, district policy opinion is positively and statistically significantly related to candidates' CPIs, indicating systematic align-

ment between candidates' issue positions and local constituency preferences.

This relationship persists even after controlling for overall district ideology (right-hand models in Figure 6). District-level policy opinion remains a statistically significant predictor of issue-specific CPI scores across all six models, though at varying levels of significance. Overall district ideology coefficients (gray points) provide a magnitude benchmark for the policy-specific opinion estimates (black points). District ideology is widely viewed as a primary determinant of candidates' issue positioning (see Ansolabehere et al. 2001; Burden 2004), making it a natural reference point for assessing the substantive importance of policy-specific district preferences. Across issue domains, the estimated effects of policy-specific district opinion are comparable in magnitude to those of overall district ideology, underscoring the substantive importance of policy-specific district preferences in candidates' issue positioning.

Moreover, district policy-specific opinion does not merely substitute for overall district ideology as a predictor of candidate positioning: ideal-point measures of district ideology remain statistically significant at the 0.01 level in four of the six models. Together, these results indicate that constituency issue-specific policy preferences and overall ideologies each make independent contributions to explaining candidates' left-right positioning. These findings suggest that candidates engage in a form of domain-specific responsiveness that operates alongside—rather than as a mere proxy for—broader ideological representation. Candidates appear to calibrate their positions not only to their district's general political leaning but also to constituency preferences on particular policy questions, consistent with candidates possessing information about local opinion that extends beyond what aggregate ideology alone would capture.

Discussion & Conclusion

This paper challenges a fundamental assumption in the study of American politics: that congressional candidates hold consistent left-right positions across issues. To advance our understanding of candidate positioning, this article introduces a novel text-to-measure pipeline for extracting and scaling latent policy positions within texts. We additionally employ an original compilation of campaign platforms sourced from congressional candidates' campaign websites to inform our

positioning estimates. We produce estimates of policy-level positioning, which we refer to as Candidate Positioning Indexes (CPIs), for three-quarters of all major-party, ballot-eligible candidates who ran in U.S. House primaries between 2018 and 2022. Our measurement focuses on major policy cleavages across six policy areas that voters expressed as pivotal to guiding their vote choice in recent elections: abortion, education, energy, guns, healthcare, and immigration. Importantly, our measurement strategy is flexible and can be adapted for various other policy applications within US politics and beyond.

Three key findings emerge from our analysis. First, we find that fewer than half of congressional candidates in our sample adopt consistently left, right, or centrist positions across issue domains, indicating that positional constraint is less prevalent than commonly assumed. On average, candidates shift 20 percentile ranks in relative left-right positioning from one issue to another—a level of variation that we show to be substantively meaningful through both qualitative and quantitative assessments. Second, this positional heterogeneity increases in accordance with strategic, district-level determinants. And finally, this variation in positioning aligns with district-level, issue-specific public opinion, suggesting that candidates strategically tailor their stances to local preferences on individual policies.

These findings appear to conflict with a large body of research demonstrating constraint in legislators' roll-call behavior. One explanation for this divergence lies in the distinct institutional settings of campaigning versus governing. Roll-call votes are shaped by strong party leadership, agenda control, and coalition maintenance, all of which exert pressure toward party-line voting and ideological conformity (Pearson 2015; Snyder and Groseclose 2000). By contrast, campaign platforms afford candidates considerable autonomy to emphasize issue positions that resonate with local constituencies, free from the formal constraints of party discipline or legislative bargaining. Work by Marble and Tyler (2022) similarly captures this disconnect, finding substantially less positional constraint and greater multidimensionality in positioning estimates based on candidate surveys than in those derived from roll-call data. Alternatively, this apparent disconnect in positional constraint between campaigns and governance may stem from fundamental differences

in the data-generating processes underlying rhetoric versus roll-call voting, or from variations in scaling methodology. Future work should disentangle these competing explanations by directly comparing the same legislators' campaign communications with their subsequent voting behavior, employing consistent measurement approaches across both domains. Additionally, examining how this campaign-governance gap varies across different electoral contexts—such as competitive versus safe districts, or primary versus general elections—could illuminate when and why legislators strategically decouple their campaign messaging from their legislative behavior.

Our findings also contribute to ongoing debates about the nature of political polarization in the United States. Rather than viewing polarization as a uniform phenomenon across all policy domains, our results suggest that the intensity of elite issue polarization may vary substantially across issue areas and electoral contexts. Competitive districts, in particular, incentivize strategic positioning that operates at the issue level rather than through blanket ideological moderation. This pattern offers a cautiously optimistic perspective on democratic responsiveness: even in an era of heightened polarization, electoral competition continues to generate meaningful incentives for substantive representation on specific policies that matter to local constituencies. The persistence of these incentives suggests that electoral institutions retain their capacity to shape candidate behavior in ways that promote representation, even as broader forces push toward partisan polarization.

Several limitations of our approach point toward other productive avenues for future research. While we examine six major issue areas, extending this analysis to additional issue domains could reveal whether patterns of strategic positioning vary across issues with different partisan valences or constituency salience. Moreover, our cross-sectional design precludes causal identification of how individual candidates adjust their positions over time or in response to changing electoral conditions. Panel data tracking the same candidates across multiple election cycles would illuminate whether issue-specific positioning reflects stable public positions or dynamic strategic adaptation. Future work should also investigate the voter-side mechanisms underlying these patterns: Do voters accurately perceive candidates' issue-specific positions, or does positional heterogeneity risk confusing voters and undermining accountability? Understanding whether the electoral benefits of

issue-level strategic positioning outweigh potential normative costs remains crucial for assessing the normative implications of our findings.

This study demonstrates that candidate positioning involves a complex calculation of partisan expectations, local preferences, and competitive pressures rather than solely ideological or opportunistic motives. By clarifying the conditions under which candidates deviate from positional constraint, we highlight ways in which democratic institutions can encourage meaningful policy representation, even in an increasingly nationalized political environment. As American democracy grapples with deepening partisan divides, understanding these mechanisms of strategic moderation becomes increasingly important for finding opportunities for policy compromise and democratic renewal.

References

- Ahler, Douglas J., and David E. Broockman. 2018. "The Delegate Paradox: Why Polarized Politicians Can Represent Citizens Best." *Journal of Politics* 80(4).
- Aldrich, John H., Jacob M. Montgomery, and David B. Sparks. 2014. "Polarization and ideology: Partisan sources of low dimensionality in scaled roll call analyse." *Politial Analysis* 22(4): 435–456.
- Ansolabehere, Stephen, and Shiro Kuriwaki. 2022. "Congressional representation: Accountability from the constituent's perspective." *American Journal of Political Science* 66(1): 123–139.
- Ansolabehere, Stephen, Jr. Snyder, James M., and III Stewart, Charles. 2001. "Candidate Positioning in U.S. House Elections." *American Journal of Political Science* 45(1).
- Bafumi, Joseph, and Michael C. Herron. 2010. "Leapfrog Representation and Extremism: A Study of American Voters and Their Members in Congress." *American Political Science Review* 104(3): 519–542.
- Barber, Michael, and Jeremy C. Pope. 2019. "Does party trump ideology? Disentangling party and ideology in America." *American Political Science Review* 113(1): 38–54.
- Barbera, Pablo. 2015. "Birds of the same feather tweet together: Bayesian ideal point estimation using Twitter data." *American Political Science Review* 23(1): 76–91.
- Bolukbasi, Tolga, Kai-Wei Chang, James Y Zou, Venkatesh Saligrama, and Adam T Kalai. 2016. "Man is to computer programmer as woman is to homemaker? debiasing word embeddings." *Advances in neural information processing systems* 29.
- Bonica, Adam. 2014. "Mapping the Ideological Marketplace." *American Journal of Political Science* 58(2): 367–386.
- Bonica, Adam, and Zhao Li. 2021. "Inferring Candidates' Issue-Specific Positions from Itemized Campaign Contributions Using Supervised Machine Learning." <https://www.dropbox.com/scl/fi/ibr2a47jb4e04kf31h4hq/Bonica-and-Li-2021.pdf>.
- Brady, David W., Hahrie Han, and Jeremy C. Pope. 2007. "Primary Elections and Candidate Ideology: Out of Step with the Primary Electorate?" *Legislative Studies Quarterly* 32(1): 79–105.
- Broockman, David E. 2016. "Approaches to studying policy representation." *Legislative Studies Quarterly* 41(1): 181–215.
- Broockman, David E., and Benjamin E. Lauderdale. 2025. "“Moderates”." *American Political Science Review* .
- Burden, Barry C. 2004. "Candidate Positioning in US Congressional Elections." *British Journal of Political Science* 34: 211–227.

- Canes-Wrone, Brandice, David W. Brady, and John F Cogan. 2002. "Out of step, out of Office: Electoral Accountability and House members' voting." *Annual Review of Political Science* 96: 127–140.
- Carson, Jamie L., Jason M. Roberts, and Rachel Porter. 2020. "Congress and the Nationalization of Congressional Elections." In *New Directions in Congressional Politics*. Routledge.
- Carson, Jamie L., Joel Sievert, and Ryan D. Williamson. 2023. *Nationalized politics: Evaluating electoral politics across time*. Oxford University Press.
- Case, Colin R. 2027. "Measuring Strategic Positioning in Congressional Elections." *Journal of Politics* .
- Caughey, Devin, and Eric Schickler. 2016. "Substance and change in congressional ideology: NOMINATE and its alternatives." *Studies in American Political Development* 30(2): 128–146.
- Colao, Jacqueline, David E. Broockman, Gregory A. Huber, and Joshua L. Kalla. 2025. "Tracing Polarization's Roots: A Panel Study of Voter Choice in Congressional Primary and General Elections."
- Converse, Philip E. 1964. "The Nature of Belief Systems in Mass Publics." *Critical Review* (1-74).
- Cox, Gary W., and Mathew D. McCubbins. 2005. *Setting the Agenda: Responsible Party Government in the U.S. House of Representatives*. Cambridge: Cambridge University Press.
- Crespin, Michael H., and David W. Rohde. 2010. "Dimensions, issues, and bills: Appropriations voting on the House floor." *Journal of Politics* 72(4): 976–989.
- Daubler, Thomas, Kenneth Benoit, Slava Mikhaylov, and Michael Laver. 2012. "Natural sentences as valid units for coded political texts." *British Journal of Political Science* 42(4): 937–951.
- de Benedictis-Kessner, Justin, Ruth Greenwood, Nicholas Stephanopoulos, and Christopher Warshaw. 2024. "TrueViews–See What People Think." <https://trueviews.org/>.
- de Marchi, Scott, Spencer Dorsey, and Michael J. Ensley. 2021. "Policy and the structure of roll call voting in the US house." *American Political Science Review* 41(2): 384–408.
- Denny, Matthew J., and Arthur Spirling. 2018. "Text Preprocessing For Unsupervised Learning: Why It Matters, When It Misleads, And What To Do About It." *Political Analysis* 26(2): 168–189.
- Downs, Anthony. 1957. "An Economic Theory of Political Action in a Democracy." *Journal of Political Economy* 65: 135–150.
- Druckman, James N., Martin J. Kifer, and Michael Parkin. 2009. "Campaign Communications in U.S. Congressional Elections." *American Political Science Review* 103(3): 343–366.
- Fenno, Richard F. 1978. *Home Style: House Members in Their Districts*. New York: Longman.
- Fiorina, Morris P. 2016. "The (Re)Nationalization of Congressional Elections." <https://www.hoover.org/research/renationalization-congressional-elections>.

- Freeder, Sean, Gabriel S. Lenz, and Shad Turney. 2022. "The importance of knowing "what goes with what": Reinterpreting the evidence on policy attitude stability." *Journal of Politics* 81(1): 274–290.
- Garg, Nikhil, Londa Schiebinger, Dan Jurafsky, and James Zou. 2018. "Word embeddings quantify 100 years of gender and ethnic stereotypes." *Proceedings of the National Academy of Sciences* 115(16): E3635–E3644.
- Garten, Justin, Joe Hoover, Kate M Johnson, Reihane Boghrati, Carol Iskiwitch, and Morteza Dehghani. 2018. "Dictionaries and distributions: Combining expert knowledge and large scale textual data content analysis: Distributed dictionary representation." *Behavior research methods* 50: 344–361.
- Gimpel, James G., Frances E. Lee, and Shanna Pearson-Merkowitz. 2008. "The check is in the mail: Interdistrict funding flows in congressional elections." *American Journal of Political Science* 52(2): 373–394.
- Grand, Gabriel, Idan Asher Blank, Francisco Pereira, and Evelina Fedorenko. 2022. "Semantic projection recovers rich human knowledge of multiple object features from word embeddings." *Nature Human Behavior* 6: 975–987.
- Grimmer, Justin, and Brandon M Stewart. 2013. "Text as data: The promise and pitfalls of automatic content analysis methods for political texts." *Political analysis* 21(3): 267–297.
- Grimmer, Justin, Margaret E. Roberts, and Brandon M. Stewart. 2022. *Text as Data: A New Framework for Machine Learning and the Social Sciences*. Princeton University Press.
- Hall, Andrew B. 2015. "What Happens When Extremists Win Primaries?" *American Political Science Review* 109(1): 18–42.
- Harden, Jeffrey J. 2015. *Multidimensional democracy: A supply and demand theory of representation in American legislatures*. Cambridge University Press.
- Heckman, James J., and James M. Snyder Jr. 1996. "Linear Probability Models of the Demand for Attributes with an Empirical Application to Estimating the Preferences of Legislators." *National Bureau of Economic Research Working Paper* .
- Henderson, John A., Geoffrey Sheagley, Stephen N. Goggin, Logan Dancey, and Alexander G. Theodoridis. 2022. "Primary divisions: How voters evaluate policy and group differences in intraparty contests." *Journal of Politics* 84(3): 1760–1776.
- Herrnson, Paul S. 1992. "Campaign Professionalism and Fundraising in Congressional Elections." *Journal of Politics* 54(3): 859–870.
- Hetherington, Marc J. 2001. "Resurgent Mass Partisanship: The Role of Elite Polarization." *American Journal of Political Science* 93(619-631).
- Hopkins, Daniel J. 2018. *The Increasingly United States: How and Why American Political Behavior Nationalized*. University of Chicago Press.

- Jennings, M Kent. 1992. "Ideological thinking among mass publics and political elites." *Public Opinion Quarterly* 56(4): 419–441.
- Jochim, Ashley E., and Bryan D. Jones. 2013. "Issue politics in a polarized congress." *Political Research Quarterly* 66(2): 352–369.
- Kinder, Donald R. 1998. "Opinion and Action in the Realm of Politics."
- Kingdon, John W. 1973. *Congressman's Voting Decisions*. Harper and Row.
- Kitagawa, Risa, and Fiona Shen-Bayh. 2024. "Measuring Political Narratives in African News Media: A Word Embeddings Approach." *Journal of Politics* .
- Kozlowski, Austin C., Matt Taddy, and James A. Evans. 2019. "The Geometry of Culture: Analyzing Meaning through Word Embeddings." *American Sociological Review* 84(5): 905–949.
- Lapinski, John, Matt Levendusky, Ken Winneg, and Kathleen Hall Jamieson. 2016. "What do citizens want from their member of congress?" *Political Research Quarterly* 69(3): 535–545.
- Lau, Jey Han, and Timothy Baldwin. 2016. "An Empirical Evaluation of doc2vec with Practical Insights into Document Embedding Generation." *1st Workshop on Representation Learning for NLP* .
- Lauderdale, Benjamin E., Chris Hanretty, and Nick Vivyan. 2018. "Decomposing Public Opinion Variation into Ideology, Idiosyncrasy, and Instability." *Journal of Politics* 80(2): 707–712.
- Layman, Geoffrey C, and Thomas M Carsey. 2002. "Party polarization and" conflict extension" in the American electorate." *American Journal of Political Science* pp. 786–802.
- Le, Quoc, and Tomas Mikolov. 2014. Distributed representations of sentences and documents. In *International conference on machine learning*. PMLR pp. 1188–1196.
- Lee, Frances E. 2016. *Insecure Majorities: Congress and the Perpetual Campaign*. Chicago, IL: University of Chicago Press.
- Levendusky, Matthew. 2009. *The Partisan Sort: How Liberals Became Democrats and Conservatives Became Republicans*. Chicago, IL: University of Chicago Press.
- Marble, William, and Matthew Tyler. 2022. "The Structure of Political Choices: Distinguishing Between Constraint and Multidimensionality." *American Journal of Political Science* 30: 328–345.
- Mayhew, David R. 1974. *Congress: The Electoral Connection*. New Haven: Yale University Press.
- McCarty, Nolan. 2016. "In defense of DW-NOMINATE." *Studies in American Political Development* 30(2): 172–184.
- Mikolov, Tomas, Kai Chen, Greg Corrado, and Jeffrey Dean. 2013. "Efficient estimation of word representations in vector space." *arXiv preprint arXiv:1301.3781* .

- Milita, Kerri, Elizabeth N. Simas, John Barry Ryan, and Yanna Krupnikov. 2017. "The effects of ambiguous rhetoric in congressional elections." *Electoral Studies* pp. 48–63.
- Montagnes, Brendan Pablo, and Jon C. Rogowski. 2015. "Testing Core Predictions of Spatial Models: Platform Moderation and Challenger Success." *Political Science Research and Methods* 3(3): 619–640.
- Moser, Scott, Abel Rodríguez, and Chelsea L. Lofland. 2021. "Multiple ideal points: Revealed preferences in different domains." *American Journal of Political Science* 29(2): 139–166.
- Moskowitz, Daniel J., Jon C. Rogowski, and James M. Snyder Jr. 2024. "Parsing Party Polarization in Congress." *Quarterly Journal of Political Science* .
- Nielson, Lindsay, and Neil Visalvanich. 2017. "Primaries and candidates: Examining the influence of primary electorates on candidate ideology." *Political Science Research and Methods* 5(2): 397–408.
- Pearson, Kathryn. 2015. *Party Discipline in the U.S. House of Representatives*. University of Michigan Press.
- Poole, Keith T. 2007. "Changing minds? Not in Congress!" *Public Choice* 131: 435–451.
- Poole, Keith T., and Howard L. Rosenthal. 2011. *Ideology and congress*. Transaction Publishers.
- Poole, Keith T., and Howard Rosenthal. 1997. *Congress: A Political-Economic History of Roll Call Voting*. Oxford University Press.
- Poole, Keith T., and R. Steven Daniels. 1985. "Ideology, Party, and Voting in the U.S. Congress, 1959–1980." *American Political Science Review* 72(2): 373–399.
- Porter, Rachel, and Sarah A. Treul. 2025. "Evaluating (In)Experience in Congressional Elections." *American Journal of Political Science* 69(1): 284–298.
- Porter, Rachel, and Tyler Steelman. 2023. "No Experience Required: Early Donations and Amateur Candidate Success in Primary Elections." *Legislative Studies Quarterly* 48(2): 455–466.
- Porter, Rachel, Colin R. Case, and Sarah A. Treul. 2025. "CampaignView, a database of policy platforms and biographical narratives for congressional candidates." *Scientific Data* 1237.
- Porter, Rachel, Sarah A. Treul, and Maura McDonald. 2024. "Changing the Dialogue: Descriptive Candidacies and Position Taking in Campaigns for the US House of Representatives." *Journal of Politics* .
- Powell, Lynda W. 1982. "Issue Representation in Congress." *The Journal of Politics* 44(3): 658–678.
- Roberts, Jason M. 2007. "The Statistical Analysis of Roll-Call Data: A Cautionary Tale." *Legislative Studies Quarterly* 32(3): 341–360.

- Roberts, Jason M., and Steven S. Smith. 2003. "Procedural Contexts, Party Strategy, and Conditional Party Voting in the U.S. House of Representatives, 1971-2000." *American Journal of Political Science* 47(April): 305–317.
- Rodriguez, Pedro L., and Arthur Spirling. 2022. "Word embeddings: What works, what doesn't, and how to tell the difference for applied research." *Journal of Politics* 84(1): 101–115.
- Rogowski, Jon C, and Patrick D Tucker. 2018. "Moderate, extreme, or both? How voters respond to ideologically unpredictable candidates." *Electoral Studies* 51: 83–92.
- Rohde, David W. 1991. *Parties and Leaders in the Postreform House*. Chicago, IL: University of Chicago Press.
- Snyder, James M., and Michael M. Ting. 2002. "An Informational Rationale for Political Parties." *American Journal of Political Science* 46(1): 90–110.
- Snyder, James M., and Timothy Groseclose. 2000. "Estimating Party Influence in Congressional Roll-Call Voting." *American Journal of Political Science* 44(2): 193–211.
- Somer-Topcu, Zeynep. 2015. "Everything to Everyone: The Electoral Consequences of the Broad-Appeal Strategy in Europe." *American Political Science Review* 59(4): 841–854.
- Stone, Walter J., and Elizabeth N. Simas. 2010. "Candidate Valence and Ideological Positions in U.S. House Elections." *American Journal of Political Science* 54(2): 371–388.
- Theriault, Sean. 2008. *Party Polarization in Congress*. Cambridge University Press.
- Thomsen, Daniel M. 2022. "Competition in Congressional Elections: Money Versus Votes." Presented at the 2020 Annual Meeting of the American Political Science Association.
- Tshitoyan, Vahe, John Dagdelen, Leigh Weston, Alexander Dunn, Ziqin Rong, Olga Kononova, Kristin A Persson, Gerbrand Ceder, and Anubhav Jain. 2019. "Unsupervised word embeddings capture latent knowledge from materials science literature." *Nature* 571(7763): 95–98.
- Warshaw, Christopher, and Chris Tausanovitch. 2022. "Subnational Ideology and Presidential Vote Estimates (v2022)." <https://doi.org/10.7910/DVN/BQKU4M>.
- Warshaw, Christopher, and Jontahan Rodden. 2012. "How Should We Measure District-Level Public Opinion on Individual Issues." *Journal of Politics* 74(1): 203–19.
- Yoshinaka, Antoine, and Christian R. Grose. 2011. "Ideological hedging in uncertain times: Inconsistent legislative representation and voter enfranchisement." *British Journal of Political Science* 41(4): 765–794.

Strategic Heterogeneity in Issue Positioning: Evidence from Congressional Campaigns (Supplementary Materials for Online Publication Only)

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A1 Data Collection

To collect text data from candidate campaign websites, we first identify all major-party candidates running for the U.S. House in 2018, 2020, and 2022 using Federal Election Commission filings and state-level election records. We then locate campaign website URLs by consulting online repositories, candidates’ social media pages, and targeted Google searches. Candidates without an identifiable website are re-checked in the days leading up to their primary election. Using this procedure, we identify campaign websites for 5,228 of 6,009 major-party, ballot-eligible candidates who ran during this period. After identifying a campaign website, we determine whether the candidate also posted a policy platform. Platform pages typically have clear titles (e.g., “Where I Stand” or “Issues”). Of the candidates with campaign websites, 82% also had a policy platform, corresponding to 75% of all major-party, ballot-eligible candidates. See Porter et al. (2025) for a discussion of predictors for campaign website missingness.

We define a platform point as the body text associated with a descriptive header corresponding to a specific issue area. Platform points may appear as sections on a single page or as separate sub-pages linked from a central platform page. Text and metadata are archived using a Qualtrics form, including candidate identifiers and background characteristics such as race, gender, prior elected experience, and FEC fundraising identifiers. When necessary, supplemental biographical information is drawn from external sources, including state legislative profiles, news coverage, Wikipedia, and Ballotpedia. To ensure consistency, campaign websites are scraped during a two-week window preceding each candidate’s primary election, as many sites are deactivated shortly thereafter. Although some campaign websites are archived by repositories such as the Wayback Machine, platform pages are often missing or inconsistently captured, particularly for primary candidates. For this reason, all text data is collected in real time.

A small share of incumbents—fewer than 10% of those running for reelection—did not maintain a campaign website with a policy platform, typically operating donation-only landing pages. For these candidates, we collect policy statements from their official House.gov websites. All such cases are flagged with a dichotomous indicator in the replication materials.

A2 Left-Right Scaling

A2.1 Coding Instructions for Paragraph Classification

This task involves reading paragraphs from congressional candidates’ campaign platforms and judging whether the text pertains to issue areas of interest. If you believe the paragraph pertains to a given issue area, mark a “1” in that issue area column in the attached Excel spreadsheet. If you do not believe the paragraph pertains to a given issue area, leave that column blank. Paragraph categorizations are not mutually exclusive; this means a single paragraph can be labeled for multiple issues of interest (e.g., one paragraph can pertain to both *Abortion* and *Healthcare*). Reference the descriptions and examples below to determine whether a paragraph pertains to a given issue:

Abortion — the ability to get an abortion

Relevant Examples: pro-life, pro-choice, “reproductive healthcare”, Planned Parenthood (if explicitly tied to abortion).

Examples specific to Planned Parenthood:

- We need to defund Planned Parenthood (**yes**)
- We need to defund Planned Parenthood because it provides abortions (**yes**)

- We need to defund Planned Parenthood because they sell fetal body parts (**no**)
- We need to support Planned Parenthood (**yes**)
- We need to support Planned Parenthood because it provides abortions (**yes**)
- We need to support Planned Parenthood because it provides breast cancer screenings (**no**)

Education — access to education, government involvement, school content

Relevant Examples: teacher salaries, school choice, homeschooling, Department of Education, local school boards, free college, college debt, Common Core/No Child Left Behind reform

Energy — resources used to produce energy Relevant Examples: fossil fuels (oil, natural gas, coal), renewables (solar, wind, etc.), “all-of-the-above” policy, energy independence; Does **not** include: discussions solely about climate change, environment, conservation

Guns — the ability to purchase weapons Relevant Examples: gun rights, limits on purchasing, 2nd amendment, red flag laws, concealed carry—requires mention of access/restriction, not just NRA or school shootings alone

Healthcare — access, cost, or quality of healthcare Relevant Examples: Obamacare, insurance, rural health, Medicare, COVID-19, vaccines; abortion included if described as “abortion as health-care”; Does **not** include: lockdowns or business closures

Immigration — entry to or treatment of immigrants in the U.S.

Relevant Examples: reform, asylum, border security, DACA, services to immigrants, deportation, ICE, family separation

Policing — role, support, or accountability of police Relevant Examples: body cameras, “back the Blue,” defunding police, brutality, shootings; Does **not** include: prison reform, school-to-prison pipeline, general criminal justice system

Voting Rights — who can vote and how Relevant Examples: voter ID, ballot fraud, audits, mail-in ballots, holiday voting, ranked-choice voting, enfranchisement (felons, Native Americans); January 6th references must be tied to election security

A2.2 Details on Machine Learning Paragraph Classifier

We train a series of supervised machine learning models to predict whether a campaign platform paragraph discusses topics related to our issues of interest. We train four separate classifiers for each of our six issue areas: support vector machine, ridge regression, random forest, and XGBoost. For all models, we select model parameters using a 5-fold cross-validation grid search. We use predictions for all four models to train a logistic regression stacking classifier. Following the recommendations of Park and Montgomery (2025), we split our 8,262 labeled campaign platform paragraphs into an 80-20 training-validation split, with 6,609 for model training and 1,653 for downstream validation. Before model fitting, we pre-process paragraphs by converting all text to lowercase and removing stop words and punctuation. For each classification model (four base models and one ensemble model), we make out-of-sample predictions for all unlabeled paragraphs, as well as the held-out set. Table A1 presents fit statistics for our classifier. Across all six issue areas of interest, our ensemble classifier achieves the highest out-of-sample precision, recall, and F1 score on our validation data.

Table A1: Out-of-Sample Validation Metrics for Stacking Classifier

	Accuracy	Precision	Recall	F1
Abortion	0.992	0.852	0.912	0.881
Education	0.963	0.803	0.893	0.846
Energy	0.987	0.832	0.944	0.884
Guns	0.992	0.904	0.955	0.929
Healthcare	0.971	0.752	0.911	0.824
Immigration	0.983	0.893	0.862	0.877

A2.3 Full List of Text Pro-Processing Steps

We take the following steps when pre-processing text from our corpus of paragraphs drawn from candidates' campaign platforms:

- All text to lowercase
- String pattern 'medicare for all' to 'medicareforall'
- String pattern 'medicare-for-all' to 'medicareforall'
- String pattern 'pro choice' to 'prochoice'
- String pattern 'pro-choice' to 'prochoice'
- String pattern 'pro-life' to 'prolife'
- String pattern 'de-escalation' to 'deescalation'
- String pattern 'dodd-frank' to 'doddfrank'
- String pattern 'k-12' to 'ktwelve'
- String pattern 'k12' to 'ktwelve'
- String pattern 'pre-k' to 'prek'
- String pattern '4-year' to 'four year'
- String pattern '4 year' to 'four year'
- String pattern '2-year' to 'two year'
- String pattern '2 year' to 'two year'
- String pattern 'non-violent' to 'nonviolent'
- String pattern '2nd amendment' to 'second amendment'
- String pattern '2a' to 'second amendment'
- String pattern 'non-profit' to 'nonprofit'
- String pattern 'non-discrimination' to 'nondiscrimination'
- Replace all '-' with ' '
- Remove all non-alphabetic characters
- Remove extraneous UTC code
- Trim white space

A2.4 Term Selection Strategy & Robustness

To determine the terms that will define the left-most and right-most positions for our eight issue cleavages, we first consulted the policy platforms of far-left and far-right advocacy groups. We additionally consult policy priorities made available by far-left and far-right congressional caucuses. For each issue area, we consulted the following resources:

Abortion: Far-Right

- Eagle Forum: <https://eagleforum.org/topics/pro-life.html>
- Family Research Council: <https://www.frc.org/abortion>

- **Heritage Foundation:** <https://www.heritage.org/life-and-family>
- **National Right to Life:** <https://nrlc.org/>
- **Susan B. Anthony:** <https://sbaprolife.org/about>

Abortion: Far-Left

- **Center for American Progress:** <https://www.americanprogress.org/topic/abortion-rights/>
- **House Progressive Caucus:** <https://progressives.house.gov/universal-health-care>
- **Justice Democrats:** <https://justicedemocrats.com/platform/society/#reproductive-rights>
- **Our Revolution:** <https://ourrevolution.com/policy-fights/>
- **Planned Parenthood:** <https://www.plannedparenthoodaction.org/issues/abortion>
- **Reproductive Freedom for All:** <https://reproductivefreedomforall.org/about/>

Education: Far-Right

- **Club for Growth:** <https://www.clubforgrowth.org/issue/education/>
- **Conservative Caucus:** <https://www.theconservativecaucus.org/about/stand-for-fight-for>
- **CPAC:** <http://ratings.conservative.org/issues?group=B>
- **Eagle Forum:** <https://eagleforum.org/topics/education.html>
- **Family Research Council:** <https://www.frc.org/education>
- **FreedomWorks:** <https://www.freedomworks.org/issue/curriculum/>
- **Heritage Foundation:** <https://www.heritage.org/empower-parents-make-education-choices>

Education: Far-Left

- **Center for American Progress:** <https://www.americanprogress.org/topic/education-k-12/>
- **Congressional Progressive Caucus Center:** <https://www.progressivecaucuscenter.org/debt-free-college>
- **House Progressive Caucus:** <https://progressives.house.gov/education>
- **Justice Democrats:** <https://justicedemocrats.com/platform/economy/#cancel-student-debt>

Energy: Far-Right

- **CPAC:** <http://ratings.conservative.org/issues?group=C>
- **Conservative Caucus:** <https://www.theconservativecaucus.org/about/stand-for-fight-for>
- **FreedomWorks:** <https://www.freedomworks.org/vote/energy-independence-and-security-act/>
- **Heritage Foundation:** <https://www.heritage.org/energy>

Energy: Far-Left

- **Center for American Progress:** <https://www.americanprogress.org/topic/clean-energy-2/>
- **Congressional Progressive Caucus Center:** <https://www.progressivecaucuscenter.org/issues/environment-climate>
- **House Progressive Caucus:** <https://progressives.house.gov/climate-justice>
- **Our Revolution:** <https://ourrevolution.com/policy-fights/>

Guns: Far-Right

- **CPAC:** <http://ratings.conservative.org/issues?group=A>
- **Gun Owners of America:** <https://www.gunowners.org/about-goal/>
- **National Association for Gun Rights:** <https://www.nationalgunrights.org/about-us/nagr-pac/>
- **National Rifle Association:** <https://home.nra.org/statements/nra-statement-on-gun-control-package/>

Guns: Far-Left

- **Center for American Progress:** <https://www.americanprogress.org/topic/gvp/>
- **Congressional Progressive Caucus Center:** <https://www.progressivecaucuscenter.org/issues/gun-violence-prevention>
- **Everytown for Gun Safety:** <https://www.everytown.org/>

- **Justice Democrats:** <https://justicedemocrats.com/platform/society/#gun-safety>
- **Moms Against Gun Violence:** <https://momsdemandaction.org/about/>

Healthcare: Far-Right

- **CPAC:** <http://ratings.conservative.org/issues?group=F>
- **Family Research Council:** <https://www.frc.org/health-care>

Healthcare: Far-Left

- **Center for American Progress:** <https://www.americanprogress.org/topic/health-coverage-and-access/>
- **House Progressive Caucus:** <https://progressives.house.gov/universal-health-care>
- **Medicare-For-All Action Network** <https://actionnetwork.org/letters/medicare-for-all-caucus/>
- **Our Revolution:** <https://ourrevolution.com/policy-fights/>
- **Congressional Progressive Caucus Center:** <https://www.progressivecaucuscenter.org/medicare-for-all>

Immigration: Far-Right

- **Conservative Caucus:** <https://www.theconservativecaucus.org/about/stand-for-fight-for>
- **Heritage Foundation:** <https://www.heritage.org/borders-and-crime>

Immigration: Far-Left

- **Center for American Progress:** <https://www.americanprogress.org/topic/immigration/>
- **Congressional Progressive Caucus Center:** <https://www.progressivecaucuscenter.org/issues/immigrants-rights>
- **House Progressive Caucus:** <https://progressives.house.gov/immigrant-rights>

We relied on these resources to produce the term dictionaries listed in Table A8 below. In each dictionary, we include at least 25 terms to ensure that our measurement results are not dependent on any particular term. Existing work demonstrates that long-term lists effectively capture the semantic meaning of a latent construct of interest. As a final step, we verify that our corpus accurately reflects the concepts we aim to capture in our positional dictionaries. We achieve this by cross-referencing our term dictionaries with the vocabulary of our corpus, and then randomly sampling documents to ensure that terms are used in expected contexts. An alternative approach to term selection could rely on a computer-assisted, statistical approach to keyword selection. When implementing the approach proposed by King et al. (2017) for computer-assisted keyword discovery, we found our term list more closely reflected our core concepts.

To demonstrate that our estimates are robust to alternative term selections, we randomly sample 15 terms from Table A11 and re-estimate CPIs. We repeat this process 50 times for each issue area. We then correlate each of these scores with the CPIs used in the body of the paper. Consistently, we find very high correlations with our measures, both across and within parties. Below are the average correlations across the 50 keyword samples for each issue area. This analysis demonstrates that CPI estimation is not sensitive to the inclusion or exclusion of specific keywords.

- **Abortion:** Cross-Party (0.970); Dems (0.904); Reps (0.930)
- **Education:** Cross-Party (0.937); Dems (0.873); Reps (0.916)
- **Energy:** Cross-Party (0.931); Dems (0.877); Reps (0.896)
- **Guns:** Cross-Party (0.971); Dems (0.922); Reps (0.943)
- **Healthcare:** Cross-Party (0.946); Dems (0.893); Reps (0.919)
- **Immigration:** Cross-Party (0.937); Dems (0.902); Reps (0.861)

Table A2: Issue Cleavages & Full Positional Term Dictionaries

Issue Area	Left-Most Position	Right-Most Position
Abortion	Pro-Choice: reproductive, justice, freedom, health, care, healthcare, expand, access, safe, hyde, roe, wade, choice, pro-choice, government, codifying, legal, restrictive, decisions, autonomy, control, privacy, doctor, respect, personal, equality, women, sexual, services	Pro-Life: life, birth, death, unborn, womb, precious, sanctity, moment, conception, heartbeat, begins, prolife, partial, fetal, born, alive, demand, saving, defund, taxpayer, infanticide, murder, barbaric, innocent, dignity, values, conscientious, immoral, vulnerable, defenseless, ban, outlaw, god, gift, sacred
Education	Federal Involvement: public, underfunded, invest, government, responsibility, cuts, investment, expand, teachers, affordable, free, tuition, debt, universal, profit, prek, loan, cancel, college, forgiveness, income, race, live, zip, accessible, equality, pay, salaries, technology, head, start	Local Control: parents, bureaucrats, burdensome, tax, state, local, board, know, best, values, control, autonomy, decisions, abolish, eliminate, dismantle, department, decentralize, choice, competition, religious, charter, private, homeschool, vouchers, mandates, core, common, curriculum
Energy	Renewables Investment: renewable, clean, green, sustainable, solar, wind, hydroelectric, transition, paris, rejoin, emissions, climate, change, environment, ban, pollution, technology, research, development, incentives, credits, tax, subsidize, funding, investment, new, deal, net, zero, hundred	Fossil Fuel Investment: oil, gas, coal, key-stone, pipeline, independence, foreign, reliance, deregulate, cut, reopen, restrictions, red, tape, bureaucratic, repeal, free, market, private, competition, companies, china, russia, middle, east, national, security, disruption, unproven, unstable, domestic, expand, production, reserves
Guns	Increase Restrictions: mandatory, background, ban, national, hate, registry, database, assault, automatic, ar, ak, military, battlefield, war, waiting, age, years, ghost, loopholes, transfer, traumatic, violence, mass, killing, crisis, epidemic, dealers, manufacturers, trafficking, comprehensive, stricter	Increase Access: second, amendment, owner, abiding, right, infringed, inherent, unconstitutional, founding, enumerated, confiscation, unrestricted, repeal, oppose, abolish, any, overturn, enshrined, reciprocity, concealed, carry, hearing, security, freedoms, fundamental, defending, families, protection
Health Insurance	Publicly Funded: universal, all, deny, coverage, privilege, guarantee, uninsured, gap, public, medicareforall, singlepayer, free, strengthen, expand, comprehensive, dental, hearing, vision, access, human, right, equality, fundamental, poor, justice, inclusive, profit, preexisting, bankruptcy	Privatize: repeal, replace, problems, flawed, free, market, open, competition, freedom, national, state, choice, consumers, individual, patient, provider, insurance, mandate, Washington, burdensome, tape, involvement, bureaucratic, deregulate, tax, hsa, savings, obamacare, socialized
Immigration	Inclusive: undocumented, pathway, roadmap, temporary, tps, dreamers, daca, asylum, humane, compassion, dignity, diverse, culture, xenophobic, racist, families, reunification, detention, cages, humanitarian, streamline, backlogs, courts, accessible, opportunity, expand, abolish, ice, education, healthcare	Exclusive: enforce, rule, law, secure, build, wall, barrier, surveillance, technology, verify, screening, safety, protect, sanctuary, chain, migration, anchor, birthright, english, skilled, merit, labor, overstay, deport, amnesty, illegal, criminal, terrorists, flooding, porous, close, funding, hire, officers, patrol

A3 Measurement Validation - Human Rating

We compare human judgments of campaign platform text to our CPIs as a benchmark for validity. Per Lowe and Benoit (2013), “valid positional estimates from quantitative scaling, for a given dimension, should match a human reader’s placement of these texts with respect to identifying relative differences along this dimension” (p. 300). Given humans’ superior ability to read and understand the meaning of natural language, directly comparing the quantitative scaling of text to qualitative human judgments offers a meaningful benchmark for semantic validity.

In this validation exercise, two student readers and one principal investigator (PI) scored the same random selection of documents ($n = 900$). Texts were provided at units of aggregation identical to those employed to generate candidate-issue embeddings. Readers scored documents on a five-point integer scale from very left (-2) to very right ($+2$). When applicable, readers flagged ambiguous or irrelevant content. Such a lack of clarity would suggest that the differences detected through scaling fail to capture intended quantities. Instructions given to human raters for this task are provided at the end of the Appendix document.

A3.1 Inter-coder Reliability Descriptive Statistics

Below, we summarize descriptive statistics from the human rating task. Table A3 reports inter-coder agreement in scoring policy platform paragraphs; for most texts, at least two readers assigned the same score. Table A4 shows the proportion of documents in each issue area flagged as containing only irrelevant text, based on issue-specific coding guidance detailed in rater instructions. On average, approximately 6% (about eighteen documents per issue area) were classified as irrelevant. Table A5 reports the proportion of documents flagged as taking an ambiguous position; raters identified ambiguity in roughly 10% of documents (about thirty per issue area).

Table A3: Reader Agreement Rate, By Issue Area

Issue Area	3/3 Agreement	2/3 Agreement	No Agreement
Abortion	0.675	0.322	0.003
Education	0.597	0.403	0.000
Energy	0.619	0.378	0.003
Guns	0.554	0.446	0.000
Healthcare	0.551	0.425	0.024
Immigration	0.486	0.510	0.003

Table A4: Flagged Irrelevant Text by Issue Area

Issue area	PI	RA1	RA2
Abortion	0.010	0.069	0.035
Education	0.095	0.159	0.081
Energy	0.058	0.041	0.037
Guns	0.014	0.007	0.014
Healthcare	0.007	0.000	0.031
Immigration	0.027	0.027	0.051

Table A5: Flagged ambiguous text by issue area

Issue area	PI	RA1	RA2
Abortion	0.017	0.021	0.028
Education	0.047	0.098	0.159
Energy	0.051	0.054	0.116
Guns	0.035	0.045	0.059
Healthcare	0.024	0.062	0.103
Immigration	0.084	0.047	0.098

A3.2 CPI-Human Judgment Correlations by Party

Figures A1 and A2 present a series of scatter plots comparing human ratings of text with their associated CPI scores. Text flagged as “ambiguous” or “irrelevant” is assigned a value of zero (i.e., centrist). Each facet corresponds to one of the six issue areas. Across issues, correlations between human judgments and estimated CPIs range from 0.74 to 0.88, comparable to those reported in prior studies assessing the convergent validity of automated scaling methods against human evaluations (see Park and Montgomery 2025 and Ruedin and Morales 2019).

Correlations based on within-party human ratings are smaller but remain substantively strong, ranging from 0.50 to 0.71. This constitutes a considerably more stringent validation test, as it requires the CPI to recover meaningful positional variation among texts that share a common partisan label and therefore exhibit substantially less dispersion on the left–right dimension. Taken together, results from this validation exercise suggest that CPIs (1) capture the intended ideological cleavages specified in Table 1 of the manuscript and (2) correspond closely with human placements of political text on a left–right spectrum.

Figure A1: CPI Validation with Human Ratings

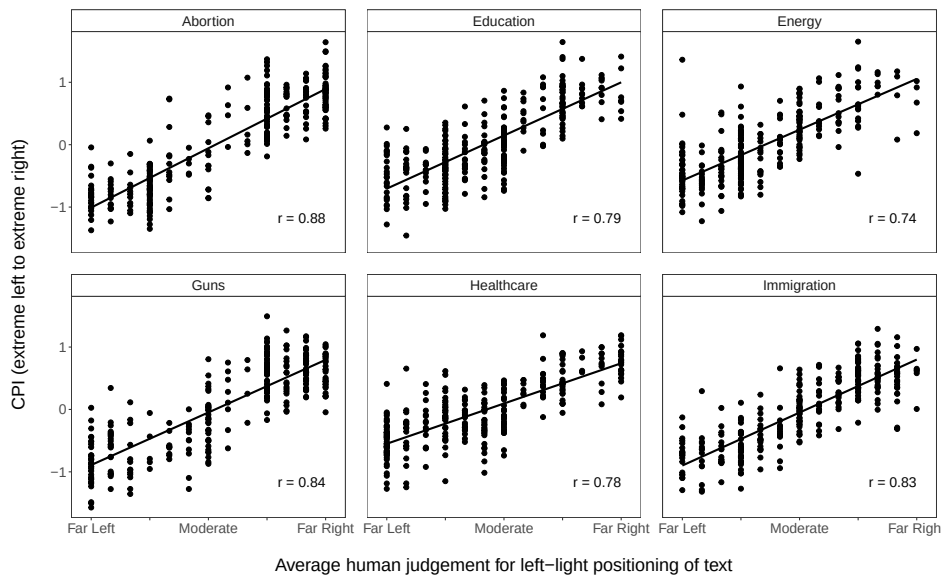
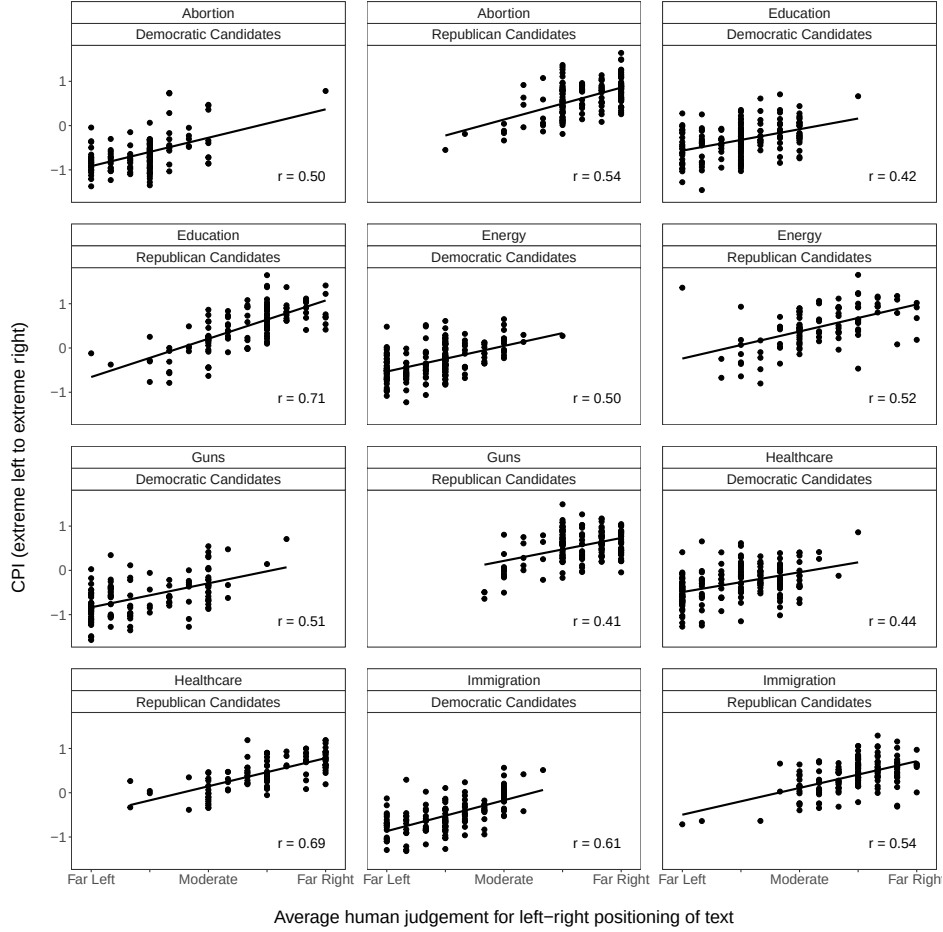


Figure A2: CPI Validation with Human Ratings, By Party

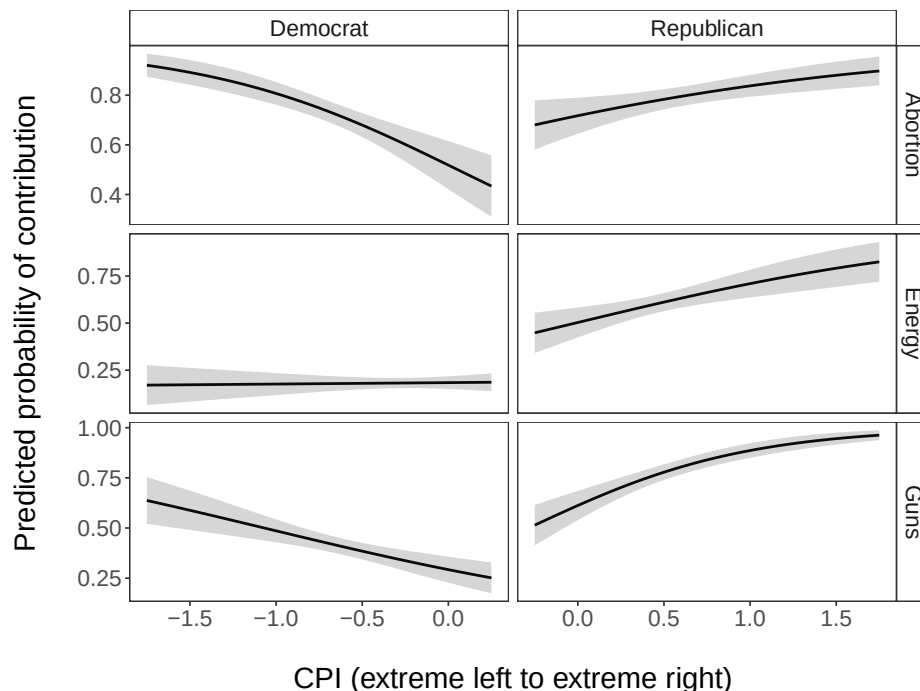


A4 Measurement Validation - PAC Giving

As a test of predictive validity, we examine the relationship between Political Action Committee (PAC) contributions and CPI extremity. Extensive research documents the giving behavior of groups, finding that these entities contribute to candidates for access-oriented and ideological reasons (e.g., Fourniaies and Hall 2014; Barber 2016; Meisels 2024). Politicians may even adjust their position-taking behavior to enhance PAC appeal (Baker 2016). For our purposes, we remain neutral regarding the direction of the causal relationship between candidate positioning and PAC contributions. The foundation of our predictive validation exercise is that candidates whose issue positions align more closely with the priorities of PAC-funded groups should receive fundraising contributions from these groups (Li 2018; Bonica and Li 2021).

A4.1 CPI-PAC Giving: Predicted Probabilities

Figure A3: CPI Validation with Position-Aligned PAC Giving



Note: Predicted probabilities for presence/absence of abortion-related PAC (top facet), energy-related PAC (middle facet), or gun-related PAC contribution as a function of CPI-Abortion, CPI-Energy or CPI-Guns.

Figure A3 presents predicted probabilities for PAC giving as a function of CPI extremity. The dependent variables across the logistic regressions are binary indicators for the presence (or absence) of a PAC donation from an abortion-related PAC (top facet), an energy-related PAC (middle facet), or a gun-related PAC (bottom facet). We identified PACs as aligned with left (right) leaning positions on abortion, energy, and guns using data from OpenSecrets. In our abortion and gun analyses, PACs constitute ideological/single-issue groups advocating for issue-specific positions. In our energy analyses, PACs constitute private companies in the energy sector. We constrain units of analysis to general election contenders, as PAC giving is relatively rare in primaries. Units include only candidates for whom we estimate CPI-Abortion, CPI-Energy, or CPI-Guns; as discussed earlier, we do not generate issue-specific CPIs for candidates with no issue-specific text. We include fundraising data for all giving in support of a candidate (i.e., direct donations and independent expenditures). Models control for unidimensional positioning, as estimated by Bonica (2024). Bivariate models produce substantively identical results and can be found in Appendix Tables A6 to A8. Full model outputs are available in Appendix Tables A6 to A8.

Across five of the six panels in Figure A3, we find a statistically significant association between PAC giving and the extremity of issue-specific candidate positioning. The relationship between energy PAC contributions and CPI is not significant for primary election winners. Recall that negative (positive) CPI values indicate more extreme left (right) issue positioning. For our models of Democratic and Republican candidates, the association between PAC giving and positional extremity is in the expected direction.

Table A6: CPI Validation with Abortion PAC Giving

	DV: Pro-Choice PAC Contribution		DV: Pro-Life PAC Contribution	
	Main Model	Bivariate	Main Model	Bivariate
CPI-Abortion	-1.357** (0.269)	-1.478** (0.265)	0.708** (0.248)	0.575* (0.242)
DIME-CFScore	3.132** (0.533)		-2.640** (0.593)	
Constant	2.771** (0.550)	-0.236 (0.184)	4.060** (0.765)	0.813** (0.174)
Observations	585	585	506	506
Party	Democrat	Democrat	Republican	Republican

Note: CPIs and CFScores range from negative (left) to positive (right). Positive (negative) coefficients indicate that as a candidate becomes more right- (left-)leaning, they are more likely to receive PAC funding. As a result, we expect negative coefficients for our Democratic models (left-leaning candidates are more likely to receive contributions) and positive coefficients for our Republican models (right-leaning candidates are more likely to receive contributions). [†]p < .10; *p < .05; **p < .01.

Table A7: CPI Validation with Energy PAC Giving

	DV: Clean Energy PAC Contribution		DV: Fossil Fuel PAC Contribution	
	Main Model	Bivariate	Main Model	Bivariate
CPI-Energy	0.052 (0.249)	0.173 (0.243)	0.881** (0.278)	0.633* (0.259)
DIME-CFScore	1.926** (0.315)		-2.466** (0.458)	
Constant	0.168 (0.269)	-1.398** (0.108)	2.936** (0.547)	0.161 (0.152)
Observations	827	827	442	442
Party	Democrat	Democrat	Republican	Republican

Note: CPIs and CFScores range from negative (left) to positive (right). Positive (negative) coefficients indicate that as a candidate becomes more right- (left-)leaning, they are more likely to receive PAC funding. As a result, we expect negative coefficients for our Democratic models (left-leaning candidates are more likely to receive contributions) and positive coefficients for our Republican models (right-leaning candidates are more likely to receive contributions). [†]p < .10; *p < .05; **p < .01.

Table A8: CPI Validation with Guns PAC Giving

	DV: Gun Control PAC Contribution		DV: Gun Rights PAC Contribution	
	Main Model	Bivariate	Main Model	Bivariate
CPI-Guns	−0.827** (0.216)	−0.856** (0.215)	1.596** (0.252)	1.281** (0.227)
DIME-CFScore	0.464 (0.342)		−3.606** (0.534)	
Constant	−0.484 (0.355)	−0.917** (0.158)	4.729** (0.675)	0.351* (0.143)
Observations	609	609	576	576
Party	Democrat	Democrat	Republican	Republican

Note: CPIs and CFScores range from negative (left) to positive (right). Positive (negative) coefficients indicate that as a candidate becomes more right- (left-)leaning, they are more likely to receive PAC funding. As a result, we expect negative coefficients for our Democratic models (left-leaning candidates are more likely to receive contributions) and positive coefficients for our Republican models (right-leaning candidates are more likely to receive contributions). [†]p < .10; *p < .05; **p < .01.

A5 Main Paper Results

A5.1 Establishing Heterogeneity in Policy-Level Positioning

Figure A4: Raw CPI Pairwise Correlations, By Candidate Type

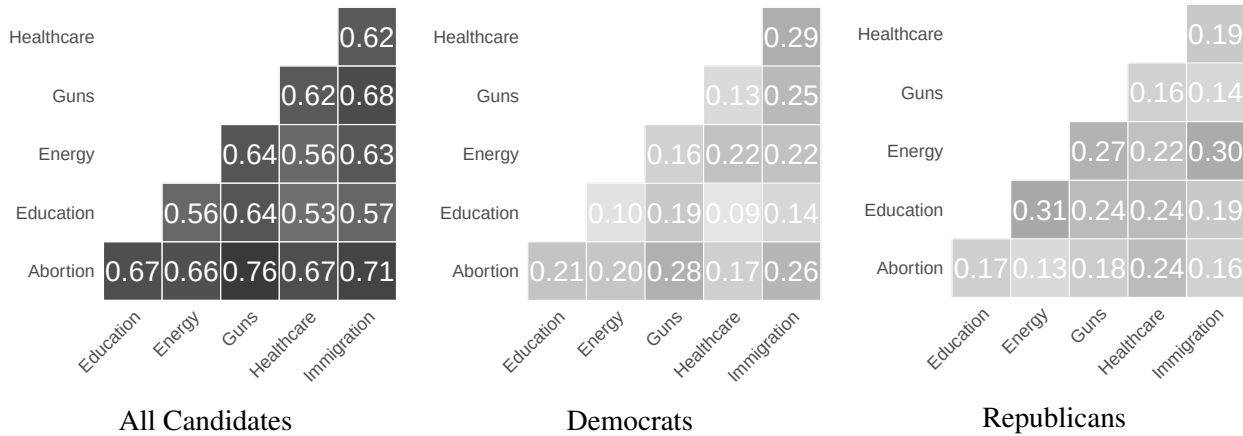


Table A9: Campaign Platform Excerpts at 20 Percentile-Rank Intervals for CPI-Abortion

Percentile	Candidate (Year)	Excerpt from Campaign Platform
10th (L)	Mike Quigley (D-IL, 2018)	“Mike trusts women to make their own healthcare choices and is a stalwart champion for women’s reproductive rights...He supports a full repeal of the Hyde Amendment and was active in the fight against anti-choice measures being included in the Affordable Care Act.”
30th	Stephen Lynch (D-MA, 2020)	“Stephen supports the Equal Rights Amendment... When Republicans tried to defund women’s health organizations, like Planned Parenthood, Stephen stood on the floor of the House of Representatives to fight the legislation.”
50th	Mark Meadows (R-NC, 2018)	“As a longtime pro-life activist, I will work in Congress to end the tragedy of abortion. I’ll work just as hard to reduce barriers to adoption, and promote policies that support groups offering hope and assistance to mothers facing crisis pregnancies.”
70th	Sean Duffy (R-WI, 2018)	“ Pro-life: Every human life has value and potential. Sean will vigorously fight to defend the right to life of the unborn and our elderly, and he will actively support policies that offer women non-violent choices and opportunities when faced with a crisis pregnancy.”
90th (R)	Bill Huizenga (R-MI, 2022)	“Protecting life beginning with conception is a value that cannot be compromised... I stressed this for 10 years as the Right to Life representative for my church in Zeeland and continue to fight for the unborn as a member of Congress.”

Table A10: Campaign Platform Excerpts at 20 Percentile-Rank Intervals for CPI-Education

Percentile	Candidate (Year)	Excerpt from Campaign Platform
10th (L)	Donald McEachin (D-VA, 2022)	“Donald will work towards universal access to pre-K education and affordable, high-quality childcare options. He will also seek overdue investments in our public schools, ensuring that teachers have the resources and support they need to close achievement gaps and best help our children.”
30th	Kathy Castor (R-FL, 2018)	“Kathy believes that every child deserves a good education and she has worked hard to ensure that each student has the chance to succeed through her work with Pell Grants, student loans, Upward Bound and Head Start. She has also worked diligently to fend off Republican cuts to students, schools...”
50th	Daniel Crenshaw (R-TX, 2020)	“It is crucial that we prepare the next generation for the workforce. We must promote vocational training as a sensible alternative to a college education and allow for Pell Grant flexibility to allow students that option...”
70th	Ben Cline (R-VA, 2022)	“I will work to ensure we put children first, not bureaucrats and unions...it is essential that we empower parents to have a leading role in directing their children’s education.”
90th (R)	Ron Estes (R-TN, 2022)	“I will never support the Federal Government taking power further away from We the People. From our second amendment rights, to the education of our children, to our right to worship as we choose, our liberties must be protected from those who would...tell us how we educate our children.”

Table A11: Campaign Platform Excerpts at 20 Percentile-Rank Intervals for CPI-Energy

Percentile	Candidate (Year)	Excerpt from Campaign Platform
10th (L)	Seth Moulton (D-MA, 2020)	“The Green New Deal needs to be built around green jobs and clean energy, decarbonization and breakthrough technologies, and access to green energy for the developing world—and America should lead the way in winning this moral and economic opportunity...”
30th	Paul Tonko (D-NY, 2022)	“I am proud to sit on the Committee for Science, Space and Technology, and the Energy and Commerce Committee. As an engineer, I understand the importance of clean energy and the continued advancement of innovation in technology sectors.”
50th	Tom Reed (R-NY, 2018)	“Tom believes that low-cost energy is another key driver of job creation and economic growth. It’s why he supports an all-of-the-above approach to our energy needs to improve our nation’s energy infrastructure, incentivize energy exploration...promote the workforce of tomorrow.”
70th	Dana Rohrabacher (R-CA, 2018)	“Dana has strongly supported efforts to make America energy self-sufficient. He stood up for expanding domestic oil and gas production, be it by fracking or other approaches. He also championed the development of new safe nuclear alternatives...fought to open federal lands to solar power projects.”
90th (R)	Virginia Foxx (R-NC, 2020)	“God blessed America with abundant resources, including fossil fuels and renewable energy sources. All of those resources should be safely developed and Washington bureaucrats should not be able to arbitrarily lock away those resources on public land.”

Table A12: Campaign Platform Excerpts at 20 Percentile-Rank Intervals for CPI-Guns

Percentile	Candidate (Year)	Excerpt from Campaign Platform
10th (L)	Alma Adams (D-NC, 2020)	“Gun control legislation is essential to fighting crime, protecting our citizens, and reducing dangers our law enforcement face each day. Congress needs to ban certain high powered assault weapons, close the gun show loophole, require background checks for purchases of firearms, and improve the tracking of firearms in our country.”
30th	John Garamendi (D-CA, 2020)	“Garamendi also supports stopping people on terrorist watch lists from purchasing arms, particularly guns designed to kill as many people as possible as quickly as possible.”
50th	Lloyd Smucker (R-PA, 2020)	“Voted for concealed carry reciprocity and NRA-supported improvements to background checks. Cosponsored Federal Firearms Licensee Protection Act (H.R. 3790), which would enhance penalties for the theft of a firearm from federally licensed firearm stores...”
70th	Brian Babin (R-TX, 2022)	“As a 30-year member of the NRA, I am a rock-solid defender of the Second Amendment...In Congress, I have actively introduced, cosponsored, and voted for important legislation that protects the gun rights of all law abiding American citizens.”
90th (R)	Doug LaMalfa (R-CA, 2020)	“I am an ardent defender of our Second Amendment rights. I believe firmly in the individual right to keep and bear arms. I will fight any attempt to water down or weaken our rights, while pushing to ensure our rights to hunt and defend ourselves are expanded. I am proud to have earned an A rating and the endorsements of the NRA and Gun Owners of America.”

Table A13: Campaign Platform Excerpts at 20 Percentile-Rank Intervals for CPI-Healthcare

Percentile	Candidate (Year)	Excerpt from Campaign Platform
10th (L)	Katherine Clark (D-CT, 2022)	“We believe access to health care should be a right, not a privilege. Katherine Clark is: An original co-sponsor of Medicare for All; expanded access to affordable care through the American Rescue Plan.”
30th	Frank Pallone (D-NJ, 2022)	“Despite the consistent efforts of the Republican majority in Congress to deny affordable health care to millions of Americans and drive up the cost of care for everyone, I remain vigilant to protect the right to health care for all. I will continue to fight to fund the Children’s Health Insurance Program and fight against any effort to privatize Medicare for our seniors.”
50th	John Rutherford (R-FL, 2022)	“Fought for lower treatment and premium costs across the board; protected pre-existing condition coverage; fought for dependents to stay on parents’ insurance until age 26; supported President Trump’s efforts to lower prescription drug prices for seniors.”
70th	Don Bacon (R-NE, 2022)	“I will continue to work towards market-based options to bring down health-care costs...We can increase options and competition by encouraging associational pools and allowing insurance companies to compete across state lines...I support protecting coverage for those with pre-existing conditions.”
90th (R)	David Kustoff (R-TN, 2022)	“I am proud we repealed the Obamacare Individual Mandate Tax. This means you are no longer forced by the government to buy insurance you don’t want or need. I will continue to work to dismantle Obamacare once and for all so we can offer the best healthcare in the world.”

Table A14: Campaign Platform Excerpts at 20 Percentile-Rank Intervals for CPI-Immigration

Percentile	Candidate (Year)	Excerpt from Campaign Platform
10th (L)	Ritchie Torres (D-NY, 2022)	“Ritchie is ready to fight for an agenda that lifts up the millions of immigrants who come to our country seeking a better life...Not only do we need a pathway to citizenship for the millions of undocumented individuals, but Ritchie also believes we must make our health care, educational, and vocational programs more inclusive...”
30th	Ed Case (D-HI, 2022)	“Support expanded legal immigration...Address today’s problems today and without handing them off to our children.”
50th	Ron Estes (R-KS, 2022)	“For decades, Washington insiders have ignored the realities on our Southern border. This is unacceptable...Our immigration process must deter criminals and terrorists from entering illegally while providing a fair, efficient process for law abiding individuals wishing to pursue the American dream.”
70th	Gary Palmer (R-AL, 2018)	“I am opposed to amnesty for illegal immigrants. I signed the FAIR Congressional Task Force’s No-Amnesty pledge to ‘oppose legislation that would grant any form of work authorization to illegal aliens.’ The first steps to any reform of America’s immigration system will have to be securing our borders and enforcing current immigration law.”
90th (R)	Jeff Duncan (R-SC, 2020)	“You will not find anyone in Congress who supports securing our borders, and enforcing our immigration laws more than Jeff Duncan. Jeff has stood with President Trump as he has fought against illegal immigration...has been a long advocate for banning sanctuary cities, requiring mandatory E-Verify, eliminating birthright citizenship, and ending chain migration.”

Table A15: Word Exclusivity Comparison: CPI-Education

Percentile Comparison	Higher Percentile More Right-Leaning	Lower Percentile More Left-Leaning
90th vs. 70th	parents, best, choice, children, core, curriculum, control, government, common, federal, parent, local, states, religious, mandates	veterans, rural, college, resources, access, youth, loan, economy, jobs, aid, violence, loans, mental, mike, healthcare
70th vs. 50th	mike, government, parents, children, choice, promotes, theory, teaching, state, teach, market, senate, god, esea, think	support, debt, billion, help, affordable, congresswoman, rate, york, college, ensure, student, priorities, title, apprenticeship
50th vs. 30th	districts, communities, schools, management, voted, choice, expansion, best, excellence, region, additionally, priorities, designed, determine, failure	loan, student, tuition, debt, island, act, borrowers, girls, forgiveness, hr, mike, eliminate, everyone, start, close
30th vs. 10th	local, girls, educational, private, emotional, mike, significantly, foster, policies, proper, cte, oppose, va, environmental, effective	debt, pre-k, former, make, kids, get, nevada, americans, employees, wage, working, need, massive, college, wealth

Table A16: Word Exclusivity Comparison: CPI-Energy

Percentile Comparison	Higher Percentile More Right-Leaning	Lower Percentile More Left-Leaning
90th vs. 70th	biden, oil, prices, america, administration, pipeline, inflation, keystone, independence, xl, resources, foreign, lower, americas, joe	clean, water, climate, environment, emissions, power, air, change, green, infrastructure, planet, low, funding, toxic, cleaner
70th vs. 50th	prices, domestic, independent, power, american, energy, oil, natural, regulations, western, production, independence, home, consider, gas	climate, change, clean, fossil, renewable, investments, science, transition, communities, greenhouse, invest, gases, action, investing, methane
50th vs. 30th	nuclear, science, opposed, alaska, low, economically, oceans, times, need, mexico, gases, exploration, going, heat, source	act, million, green, h.r, percent, plan, renewable, help, introduced, thats, lands, member, sustainable, create, spending
30th vs. 10th	oil, american, commerce, care, nations, congressman, island, health, ive, education, spill, district, significant, come, member	100, agriculture, carbon, subsidies, deal, 2030, green, transportation, clean, transit, caused, oregon, bold, emissions, climate

Table A17: Word Exclusivity Comparison: CPI-Guns

Percentile Comparison	Higher Percentile More Right-Leaning	Lower Percentile More Left-Leaning
90th vs. 70th	right, free, constitution, shall, second, washington, amendment, rights, stand, speech, freedoms, government, protect, fathers, well	need, guns, h.r, firearms, gun, violence, violent, bad, regulations, safety, background, act, law, solutions, mass
70th vs. 50th	amendment, right, rights, defend, bear, arms, 2nd, constitutional, second, fight, member, shall, infringed, carry, h.r	violence, checks, gun, background, school, weapons, sense, can, common, assault, common-sense, safety, public, loophole, money
50th vs. 30th	right, bear, arms, rights, citizens, amendment, 2nd, believe, second, government, law-abiding, constitution, control, protect, crime	violence, gun, ban, background, universal, loopholes, research, checks, epidemic, weapons, congress, prevention, domestic, communities, assault
30th vs. 10th	amendment, second, right, rights, bear, owner, constitution, understand, constitutional, arms, bad, border, 2nd, citizens, responsible	ammunition, mike, weapons, require, high-capacity, programs, sexual, assault, funding, prayers, congressman, military-style, streets, means, federal

Table A18: Word Exclusivity Comparison: CPI-Healthcare

Percentile Comparison	Higher Percentile More Right-Leaning	Lower Percentile More Left-Leaning
90th vs. 70th	obamacare, lines, government, competition, mandates, state, repeal, across, regulations, malpractice, individual, market, purchase, insurance, accounts	medicare, social, security, prescription, benefits, act, seniors, everyone, affordable, negotiate, drug, protect, san, drugs, georgia
70th vs. 50th	government, obamacare, san, bureaucrats, georgia, innovation, price, budget, fraud, solutions, debt, takeover, approach, balance, voting	option, women, access, reproductive, public, breast, every, longer, nursing, medicaid, including, child, texas, cancer, nations
50th vs. 30th	consumers, reduction, va, centers, 2015, telehealth, patients, markets, transparency, ways, step, together, estimated, immigrants, rules	single-payer, right, addiction, massachusetts, human, fighting, dr, universal, health, dental, without, jobs, substance, wages, interests
30th vs. 10th	hr, repeal, massachusetts, want, tax, line, plans, wages, co-sponsored, effective, republicans, effort, subsidies, let, military	mental, universal, rights, york, new, healthcare, services, dental, caucus, reproductive, invest, vision, community-based, gender, expand

Table A19: Word Exclusivity Comparison: CPI-Immigration

Percentile Comparison	Higher Percentile More Right-Leaning	Lower Percentile More Left-Leaning
90th vs. 70th	border, biden, funded, dhs, security, miles, sanctuary, federal, china, borders, 5, alien, drug, trafficking, cities	want, system, united, opportunity, american, dream, reform, become, best, politicians, economy, agriculture, process, offer, interests
70th vs. 50th	aliens, illegal, border, funding, secure, voted, year, nations, cities, government, wall, u.s, federal, fentanyl, governments	comprehensive, pathway, residency, undocumented, path, came, solution, citizenship, violence, ban, international, process, refugees, dreamers, documentation
50th vs. 30th	illegal, border, illegally, drug, president, law, laws, borders, biden, cartels, safety, southern, trafficking, executive, drugs	daca, act, children, u.s, undocumented, citizenship, dreamers, dream, detention, families, parents, pathway, workers, passing, pass
30th vs. 10th	illegal, technology, law, can, borders, ability, security, tax, small, infrastructure, fact, secure, citizens, border, illegally	detention, families, fight, end, deferred, asylum, undocumented, immigrant, ice, status, separation, congress, inhumane, fear, dignity

A5.2 District Policy Opinion & Candidate Policy-Level Positioning

Table A20: Relationship Between CPI and District Policy-Specific Opinion, Full Models

	Dependent variable: Candidate Positioning Index (CPI)					
	Abortion	Education	Energy	Guns	Healthcare	Immigration
District Opinion	0.0976*** (0.0120)	0.0347*** (0.0125)	0.1391*** (0.0145)	0.1484*** (0.0122)	0.1073*** (0.0125)	0.1461*** (0.0127)
Republican	1.558*** (0.0247)	1.349*** (0.0259)	1.312*** (0.0310)	1.468*** (0.0254)	1.324*** (0.0259)	1.351*** (0.0261)
Experienced Challenger	-0.0022 (0.0338)	0.1006** (0.0392)	0.0399 (0.0444)	0.0857** (0.0362)	-0.0255 (0.0386)	-0.0449 (0.0376)
Incumbent	-0.0740*** (0.0286)	-0.0947*** (0.0316)	0.0819** (0.0338)	0.0201 (0.0296)	0.2283*** (0.0296)	0.0104 (0.0307)
Female	-0.0924*** (0.0250)	0.0170 (0.0272)	-0.0227 (0.0306)	-0.0444* (0.0264)	-0.1228*** (0.0266)	-0.1197*** (0.0276)
Open	-0.0477* (0.0273)	-0.0506 (0.0312)	-0.0611* (0.0356)	-0.0312 (0.0287)	0.0037 (0.0307)	-0.0082 (0.0303)
2020	0.0354 (0.0280)	-0.1408*** (0.0309)	-0.1949*** (0.0349)	0.0065 (0.0284)	-0.1875*** (0.0284)	-0.1272*** (0.0299)
2022	-0.0046 (0.0276)	0.0027 (0.0306)	0.0999*** (0.0342)	0.0678** (0.0292)	-0.3007*** (0.0305)	-0.0192 (0.0304)
Constant	-0.7393*** (0.0269)	-0.5163*** (0.0281)	-0.4575*** (0.0310)	-0.8065*** (0.0274)	-0.3885*** (0.0262)	-0.6608*** (0.0289)
Observations	2,503	3,536	2,606	2,688	3,401	3,126

Note:

* p<0.1; ** p<0.05; *** p<0.01

Table A21: Relationship Between CPI and District Policy-Specific Opinion, Controlling for District Ideological Orientation

	Dependent variable: Candidate Positioning Index (CPI)					
	Abortion	Education	Energy	Guns	Healthcare	Immigration
District Opinion	0.0391* (0.0237)	0.0285** (0.0125)	0.0814* (0.0462)	0.0483** (0.0209)	0.0774** (0.0353)	0.0482* (0.0284)
District Ideology (AIP)	0.0714*** (0.0245)	0.0735*** (0.0130)	0.0621 (0.0470)	0.1317*** (0.0215)	0.0337 (0.0358)	0.1107*** (0.0290)
Republican	1.545*** (0.0250)	1.328*** (0.0262)	1.309*** (0.0312)	1.439*** (0.0256)	1.320*** (0.0260)	1.342*** (0.0263)
Experienced Challenger	-0.0037 (0.0339)	0.1027*** (0.0393)	0.0437 (0.0446)	0.0950*** (0.0362)	-0.0212 (0.0388)	-0.0384 (0.0378)
Incumbent	-0.0721** (0.0287)	-0.0874*** (0.0317)	0.0811** (0.0338)	0.0213 (0.0295)	0.2323*** (0.0296)	0.0127 (0.0308)
Female	-0.0931*** (0.0251)	0.0222 (0.0273)	-0.0211 (0.0307)	-0.0383 (0.0263)	-0.1241*** (0.0268)	-0.1170*** (0.0277)
Open Seat	-0.0483* (0.0274)	-0.0556* (0.0313)	-0.0529 (0.0357)	-0.0361 (0.0286)	0.0089 (0.0308)	-0.0148 (0.0305)
2020	0.0408 (0.0282)	-0.1287*** (0.0310)	-0.2028*** (0.0350)	0.0201 (0.0285)	-0.1880*** (0.0286)	-0.1290*** (0.0300)
2022	0.0030 (0.0277)	0.0215 (0.0306)	0.1025*** (0.0341)	0.0894*** (0.0292)	-0.3024*** (0.0305)	-0.0209 (0.0304)
Constant	-0.7428*** (0.0270)	-0.5141*** (0.0281)	-0.4549*** (0.0311)	-0.8122*** (0.0273)	-0.3886*** (0.0263)	-0.6528*** (0.0290)
Observations	2,482	3,509	2,586	2,668	3,372	3,104

Note:

*p<0.1; **p<0.05; ***p<0.01

References

- Baker, Anne E. 2016. "Do Interest Group Endorsements Cue Individual Contributions to House Candidates?" *American Politics Research* 44(2).
- Barber, Michael. 2016. "Donation Motivations: Testing Theories of Access and Ideology." *Political Research Quarterly* 69(1): 148–159.
- Bonica, Adam. 2024. "Database on Ideology, Money in Politics, and Elections: Public version 4.0 [Computer file].".
- Bonica, Adam, and Zhao Li. 2021. "Inferring Candidates' Issue-Specific Positions from Itemized Campaign Contributions Using Supervised Machine Learning." <https://www.dropbox.com/scl/fi/ibr2a47jb4e04kf31h4hq/Bonica-and-Li-2021.pdf>.
- Fouirnaies, Alexander, and Andrew B. Hall. 2014. "The Financial Incumbency Advantage: Causes and Consequences." *Journal of Politics* 76(3): 711–724.
- King, Gary, Patrick Lam, and Margaret E Roberts. 2017. "Computer-assisted keyword and document set discovery from unstructured text." *American Journal of Political Science* 61(4): 971–988.
- Li, Zhao. 2018. "How Internal Constraints Shape Interest Group Activities: Evidence from Access-Seeking PACs." *American Political Science Review* 112(4): 792–808.
- Lowe, Will, and Kenneth Benoit. 2013. "Validating Estimates of Latent Traits from Textual Data Using Human Judgment as a Benchmark." *British Journal of Political Science* 21: 298–313.
- Meisels, Mellissa. 2024. "Campaign Agendas and Issue Group Strategy in Congressional Primaries.".
- Park, Ju Yeon, and Jacob M Montgomery. 2025. "Toward a framework for creating trustworthy measures with supervised machine learning for text." *Political Science Research and Methods* pp. 1–17.
- Porter, Rachel, Colin R. Case, and Sarah A. Treul. 2025. "CampaignView, a database of policy platforms and biographical narratives for congressional candidates." *Scientific Data* 1237.
- Ruedin, Didier, and Laura Morales. 2019. "Estimating party positions on immigration: Assessing the reliability and validity of different methods." *Party Politics* 3(25): 303–314.